ELECTRIC VEHICLE MANAGEMENT SYSTEM USING ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

INDIVIDUAL THESIS

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ABSTRACT

Electric vehicle (EV) users often need reliable and understandable information about charging, battery health, maintenance, and driving efficiency. However, accessing relevant information can require searching multiple sources, while general-purpose language models may generate answers that are not grounded in verified EV information. This research component addresses the need for an accessible, context-aware conversational assistant that provides EV-related guidance using relevant knowledge.

The aim of this component is to develop an AI-Powered EV Virtual Assistant using Retrieval-Augmented Generation (RAG), a large language model (LLM), and voice support. A design-and-development approach was followed, involving EV knowledge-base preparation, RAG workflow design, prompt engineering, LLM integration, voice interaction development, and prototype testing. The proposed workflow retrieves relevant information from the knowledge base and provides it as context to the LLM to generate understandable responses to users' questions through text or voice interaction.

The prototype development focused on answering EV-related questions and providing guidance on charging practices, battery care, maintenance, and driving efficiency. Preliminary user feedback indicated that the assistant was easy to use and offered quick responses and practical recommendations. However, the response quality depends on the completeness and accuracy of the knowledge base, while voice interaction may be affected by background noise and speech recognition errors. Further systematic testing is required to quantify response accuracy, response time, and usability.

The individual contribution involves designing and developing the RAG-based virtual assistant, focusing on knowledge retrieval, context-aware response generation, prompt refinement, voice support, and component testing. The component was revised from the original Predictive Charging and Breakdown Assistance System following proposal feedback. Overall, the research demonstrates a practical approach to improving access to EV-related information through conversational AI.

Keywords: Electric Vehicles (EVs), Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Voice Interaction, Conversational AI

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
EV	Electric Vehicle
FAQ	Frequently Asked Questions
gTTS	Google Text-to-Speech
LLM	Large Language Model
NLP	Natural Language Processing
PDF	Portable Document Format
RAG	Retrieval-Augmented Generation
STT	Speech-to-Text
TTS	Text-to-Speech
UI	User Interface
UX	User Experience
VUI	Voice User Interface

1 INTRODUCTION

1.1 Overall project context

Electric Vehicles (EVs) are gaining significant attention in the field of sustainable mobility due to their potential to lower reliance on fossil fuels and increase energy efficiency [1]. But battery degradation is one of the biggest challenges as it impacts vehicle performance, driving range, maintenance costs, and battery life.

There are many reasons why lithium-ion batteries are used in EVs, among them being high energy density and high efficiency. These batteries do lose their capacity over time due to ageing, charging, temperature extremes and operating conditions [2]. Hence, prediction of battery health is significant to enhance long term battery management.

In response, the overall research project "Electric Vehicle Management System Using Artificial Intelligence and Machine Learning" was designed to create an intelligent EV decision support system that overcomes the challenges. It is composed of four major parts:

I. **AI-Based EV Operational Anomaly Detection:** Detects unusual operating conditions of a vehicle.
II. **Adaptive Battery Health Prediction and Personalized Recommendation:** Performs battery health prediction and offers battery maintenance suggestions.
III. **Intelligent EV Charging Optimization and Range Prediction:** Enables charging decisions, driving range prediction.
IV. **AI-Based EV Virtual Assistant Using Retrieval-Augmented Generation (RAG) with Voice Support**: Gives intelligent support concerning EV usage and upkeep.

These components work together with an EV management application to deliver predictive data and solutions for vehicle management.

1.2 Individual component context

The individual research component, “AI-Powered EV Virtual Assistant using Retrieval-Augmented Generation (RAG),” is part of the main research project titled “Electric Vehicle Management System Using Artificial Intelligence and Machine Learning.” This component focuses on providing users with an accessible and intelligent way to obtain information about electric vehicles, including charging practices, battery care, maintenance, and driving efficiency. It combines Retrieval-Augmented Generation (RAG) with a Large Language Model (LLM) to retrieve relevant information from a curated EV knowledge base and generate context-aware responses to user queries. In addition, voice interaction is incorporated to support a more convenient and user-friendly conversational experience. The component aims to improve access to relevant EV information through natural-language interaction while contributing to the overall functionality of the EV Management System.

1.3 Problem statement

Electric vehicle (EV) users require reliable and understandable information about charging practices, battery care, maintenance, and driving efficiency. However, finding relevant information may require users to consult multiple sources, making it difficult to obtain clear and contextually relevant answers to specific questions. Although general-purpose large language models (LLMs) can generate natural-language responses, their answers may not always be grounded in verified EV-specific information. This creates a need for an intelligent solution that can retrieve relevant information from a dedicated knowledge base and use it to generate informative responses. Therefore, this research addresses the problem of providing accessible, context-aware EV information through an AI-powered virtual assistant that combines Retrieval-Augmented Generation (RAG), an LLM, and voice interaction.

1.4 esearch gap

Existing approaches to accessing EV-related information may depend on static documentation, conventional search methods, or general-purpose conversational

AI systems. These approaches may not consistently provide users with relevant, context-specific answers through a convenient conversational interface. General-purpose LLMs may also generate responses that are not sufficiently grounded in a reliable EV knowledge source. Therefore, there is a need to explore the integration of RAG and LLM technology to retrieve relevant EV information before generating responses, together with voice interaction to improve accessibility and convenience. This research addresses this gap by developing an EV-focused virtual assistant that combines knowledge retrieval, context-aware response generation, and voice-based interaction within a single conversational system.

1.5 Aim and objectives

1.5.1 Aim

The aim of this research is to develop an AI-powered EV virtual assistant using Retrieval-Augmented Generation (RAG), a Large Language Model (LLM), and voice interaction to provide users with relevant and understandable information about electric vehicles.

1.5.2 Objectives

The specific objectives of this research are:

1. To develop a curated knowledge base containing relevant information about electric vehicles, including charging, battery care, maintenance, and driving efficiency.
2. To design and implement a RAG pipeline that retrieves relevant information from the EV knowledge base in response to user queries.
3. To integrate an LLM to generate natural-language responses grounded in the retrieved information.
4. To implement prompt engineering and conversational session management to support relevant and context-aware interactions.
5. To incorporate voice interaction to enable users to communicate with the virtual assistant through spoken queries and responses.
6. To evaluate the developed prototype in terms of response relevance, understandability, usability, and response time.

1.6 Research questions or hypotheses

This research is guided by the following research questions:

- RQ1: How can Retrieval-Augmented Generation be used to retrieve relevant EV-related information and support accurate responses to user queries?
- RQ2: How effectively can an LLM generate understandable, context-aware responses using information retrieved from an EV knowledge base?
- RQ3: How can voice interaction improve the accessibility and convenience of communicating with an EV virtual assistant?
- RQ4: How can the developed virtual assistant be evaluated in terms of response relevance, usability, understandability, and response time?

These questions guide the design, development, and evaluation of the proposed virtual assistant. They can be investigated through prototype testing and user feedback. Formal hypotheses are not necessary unless your research methodology requires statistical hypothesis testing.

1.7 Scope limitations and assumptions

Scope

This research focuses on the development of an AI-powered virtual assistant for providing EV-related information through natural-language interaction. The system uses a curated EV knowledge base, a RAG pipeline, and an LLM to generate responses to user queries. The knowledge base covers topics such as charging practices, battery care, vehicle maintenance, and driving efficiency. The component also considers voice interaction and conversational context to support a more convenient user experience. The research focuses on the virtual assistant as an individual component of the wider Electric Vehicle Management System.

Limitations

The quality and relevance of generated responses depend on the completeness, accuracy, and currency of the EV knowledge base. The system may provide incomplete or less relevant answers when the required information is unavailable in the retrieved context. The performance of the LLM may also vary depending on the query and the quality of the retrieved information. Furthermore, voice interaction may be affected by background noise, pronunciation differences, and speech-recognition errors. The system should not be assumed to provide live vehicle diagnostics, real-time battery measurements, or current charging-station availability unless these capabilities have been implemented and verified.

Assumptions

The research assumes that users can access the virtual assistant through its intended interface and submit understandable text or voice queries. It also assumes that the curated EV knowledge base contains sufficiently relevant information to address the intended questions and that the integrated language model can process the retrieved context. Where external services are required for language generation or voice processing, their availability is assumed during system operation.

1.8 Individual contribution

My individual contribution to the overall research project, "Electric Vehicle Management System Using Artificial Intelligence and Machine Learning," is the design and development of the AI-Powered EV Virtual Assistant using RAG. My work focuses on preparing the EV knowledge base, developing the RAG workflow, integrating an LLM for context-aware response generation, refining prompts, and supporting conversational session management. The component also includes voice interaction to make EV-related information accessible through natural-language communication. In addition, my individual work involves testing the assistant using EV-related questions and examining the

relevance and understandability of its responses. This component contributes a conversational AI capability to the overall EV Management System by helping users access relevant information about charging, battery care, maintenance, and driving efficiency. The scope of my contribution is limited to the virtual assistant and its integration with the overall system.

1.9 Thesis structure

This thesis is organised into chapters to present the research background, development process, and evaluation of the AI-powered EV virtual assistant.

- Chapter 1 – Introduction: Introduces the research background, individual component context, problem statement, research gap, aim and objectives, research questions, scope, limitations, assumptions, and individual contribution.
- Chapter 2 – Literature Review: Reviews existing research on EV information systems, conversational AI, large language models, Retrieval-Augmented Generation, and voice-based virtual assistants. It also identifies the research gap addressed by this study.
- Chapter 3 – Research Methodology: Describes the research approach, system requirements, knowledge-base preparation, RAG workflow, LLM integration, prompt engineering, voice interaction, and evaluation methods.
- Chapter 4 – System Design and Implementation: Presents the proposed architecture, system components, retrieval and response-generation processes, interface design, and implementation details.
- Chapter 5 – Results and Evaluation: Presents the results of prototype testing and discusses response relevance, understandability, usability, and response time based on the evidence collected.
- Chapter 6 – Discussion: Interprets the findings in relation to the research objectives, discusses the contribution of the component, and identifies limitations and challenges encountered during development and integration.
- Chapter 7 – Conclusion and Future Work: Summarises the research, reviews the extent to which the objectives were achieved, and presents recommendations for future improvements.

2 LITERATURE REVIEW AND RELATED WORK

This chapter reviews the theoretical foundations, related research, existing systems, and technologies relevant to the individual research component, AI-Powered EV Virtual Assistant using Retrieval-Augmented Generation (RAG). It focuses on how conversational artificial intelligence, Large Language Models (LLMs), information retrieval, RAG, and voice interaction can be combined to provide users with relevant information about electric vehicles (EVs). The chapter also identifies the research gap and presents the conceptual framework guiding the design and development of the proposed virtual assistant.

The literature review should establish the basis for the system design and research methodology. Claims about the strengths and limitations of previous studies should be supported by verified academic sources, rather than assumptions about existing technologies.

2.1 Review method and scope

The literature review focuses on research and technical publications related to conversational AI, LLMs, RAG-based question-answering systems, knowledge retrieval, and voice-enabled virtual assistants. Sources relevant to EV information systems, including charging practices, battery care, vehicle maintenance, and driving efficiency, are also within the scope of the review.

Relevant literature can be identified through academic databases and scholarly search platforms such as Google Scholar, IEEE Xplore, ACM Digital Library, ScienceDirect, and SpringerLink. Search terms may include “Retrieval-Augmented Generation,” “RAG-based question answering,” “large language models for conversational AI,” “voice-enabled virtual assistants,” and “AI applications in electric vehicles.” Studies should be selected according to their relevance to the research objectives, methodological clarity, publication quality, and contribution to understanding the technologies used in the proposed system.

The review examines the capabilities and limitations of existing approaches, particularly their ability to retrieve relevant information, generate understandable responses, maintain conversational context, and support voice-based interaction. It also considers how these technologies can be applied to EV-related information retrieval. The findings from the selected literature will inform the design of the knowledge base, retrieval pipeline, response-generation process, and evaluation criteria of the proposed virtual assistant.

Before finalising this section, document the databases searched, search dates, exact search terms, inclusion and exclusion criteria, and number of studies selected. These details should reflect the review you actually conducted.

2.2 Domain and theoretical foundations

2.2.1 Artificial Intelligence and Conversational AI

Artificial Intelligence (AI) refers to computational techniques that enable systems to perform tasks associated with human intelligence, including language understanding, information processing, and decision support. Conversational AI applies these capabilities to systems that interact with users through natural language. In the context of this research, conversational AI provides the foundation for allowing EV users to ask questions in their own words and receive understandable responses through a virtual assistant.

Traditional information-retrieval systems generally return documents or search results that users must interpret themselves. Conversational systems, in contrast, can use language models to formulate responses in a more natural and interactive manner. However, the usefulness of a conversational assistant depends on the relevance of the information available to it and its ability to generate responses that accurately reflect that information.

2.2.2 Large Language Models

Large Language Models (LLMs) are AI models trained on large collections of text to understand and generate human language. They can interpret user queries,

summarise information, and produce responses in natural language. These capabilities make LLMs suitable for conversational applications such as an EV virtual assistant.

Nevertheless, LLMs have limitations, including the possibility of generating plausible but incorrect information, relying on knowledge that may be outdated, and providing responses that are not supported by a specific information source. Gao et al. (2024) identify hallucination, outdated knowledge, and limited transparency as important challenges associated with LLMs.

These limitations are particularly relevant to domain-specific applications in which users expect clear and dependable information.

In the proposed system, the LLM is intended to generate natural-language answers using information retrieved from the EV knowledge base. Its role is therefore to formulate an understandable response from the supplied context rather than rely exclusively on its pretrained knowledge.

2.2.3 Retrieval-Augmented Generation

Retrieval-Augmented Generation is an approach that combines information retrieval with text generation. Lewis et al. (2020) introduced a RAG framework that combines a language model with an external non-parametric knowledge source to support knowledge-intensive natural-language tasks.

A typical RAG workflow includes the following stages:

1. Query processing: The user's question is received and prepared for retrieval.
2. Knowledge retrieval: The system searches an external knowledge source for information relevant to the query.
3. Context preparation: The retrieved information is organised into context that can be supplied to the language model.
4. Response generation: The LLM uses the query and retrieved context to generate a natural-language response.

5. Response delivery: The generated answer is returned to the user through the available interface.

RAG allows a system to incorporate domain-specific information without relying exclusively on the language model's internal knowledge. However, its performance depends on the quality of the source documents, retrieval process, and generated response. If relevant information is absent or not retrieved, the answer may remain incomplete or incorrect. Gao et al. (2024) discuss different RAG approaches and challenges associated with retrieval and generation.

For the proposed EV virtual assistant, RAG is the central approach for connecting EV-related questions with information in a curated knowledge base. The retrieved context is provided to the LLM to support responses concerning topics such as charging practices, battery care, maintenance, and driving efficiency.

2.2.4 Knowledge Bases and Information Retrieval

A knowledge base is an organised collection of information that a system can use to answer questions within a defined domain. For an EV virtual assistant, the knowledge base may contain technical guidance, frequently asked questions, maintenance information, and recommendations concerning EV use.

Information retrieval is the process of identifying information relevant to a user's query. The retrieval method must be able to identify relevant content even when the wording of the question differs from the wording used in the source documents. Depending on the implemented architecture, retrieval may use keyword matching, semantic similarity, vector embeddings, or a combination of these techniques.

The quality of retrieval influences the quality of the final response because the LLM depends on the supplied context to formulate its answer. Consequently, the knowledge base should be relevant to the intended domain, clearly organised, and reviewed for accuracy. The specific retrieval technique discussed in the final thesis should match the method actually implemented in the system.

2.2.5 Voice Interaction and Human–Computer Interaction

Voice interaction allows users to communicate with a system through spoken language instead of relying exclusively on typed input. A voice-enabled assistant commonly uses speech recognition to convert spoken input into text and may use text-to-speech technology to deliver generated responses audibly.

Voice interfaces can provide a convenient alternative to keyboard-based interaction. However, their usability can be affected by background noise, speech-recognition errors, language differences, and the design of the conversational experience. Deshmukh and Chalmeta (2024) review research on voice-user-interface usability and identify user experience and evaluation as important areas of investigation.

For this research, voice interaction complements the RAG-based question-answering process. When voice input is supported, the user's spoken question is converted into text and passed through the same retrieval and response-generation workflow used for text queries. The response can then be displayed as text and, if implemented, converted into speech.

2.2.6 Application to Electric Vehicle Information

Electric vehicle users may require information about charging procedures, battery maintenance, vehicle care, and efficient driving practices. Such information can be distributed across manuals, technical documents, and online resources. A domain-specific conversational assistant can provide an alternative way of accessing this information by allowing users to ask questions directly.

The proposed system applies conversational AI, RAG, and voice interaction to this domain. The knowledge base supplies EV-related information, the retrieval component identifies relevant context, and the LLM generates a response based on the retrieved material. Voice interaction provides an additional input and output method where supported by the implementation.

The purpose of this component is to improve the accessibility of EV information through a conversational interface. It is not assumed to replace professional vehicle servicing, provide live vehicle diagnostics, or deliver real-time charging-station data unless those capabilities are separately implemented and validated.

2.3 Related research

2.3.1 Research on Retrieval-Augmented Generation

Lewis et al. (2020) presented Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, establishing a framework that combines retrieval from an external knowledge source with language generation. This research provides a theoretical foundation for systems that need to generate responses using information beyond a model's internal parameters.

Gao et al. (2024), in Retrieval-Augmented Generation for Large Language Models: A Survey, examine the development of RAG methods and discuss challenges associated with LLM hallucination, outdated knowledge, and the use of external information. Their review describes RAG as a way to combine language generation with information retrieved from external sources.

These studies are relevant to the EV virtual assistant because they explain why combining retrieval with language generation can be useful for domain-specific question answering. However, the effectiveness of this approach in the proposed EV application must be established through testing of the implemented knowledge base, retrieval process, and generated responses.

2.3.2 Research on Voice-Enabled Virtual Assistants

Research on voice-user interfaces examines how users interact with conversational systems and how usability can be assessed. Deshmukh and Chalmeta (2024) provide a systematic review of user experience and usability research related to voice-user interfaces. Their work highlights the importance of evaluating voice-based interaction rather than assuming that speech input automatically improves the user experience.

Similarly, A Systematic Review of Voice Assistant Usability: An ISO 9241–11 Approach reviews methods used to evaluate voice assistants, including usability measures and the applicability of an established usability framework.

These studies suggest that voice support should be evaluated according to the needs of the intended users and the tasks they perform. For the proposed system, relevant evaluation areas include whether users can submit questions successfully, understand the generated answers, and interact with the assistant conveniently. Speech-recognition performance should also be considered if voice input is part of the implemented prototype.

2.3.3 Research on Domain-Specific Conversational Systems

Domain-specific conversational systems are designed to answer questions within a particular subject area by using specialised information, predefined knowledge, or external information sources. Compared with general-purpose conversational systems, a domain-specific design can make it easier to organise the information required for a defined set of user needs.

RAG provides a possible architecture for this purpose because the system can retrieve information from a selected knowledge source and supply it to an LLM during response generation. Research on RAG therefore provides a basis for exploring conversational question answering in specialised domains.

For the EV domain, a specialised assistant could organise information around charging, battery care, maintenance, and driving efficiency. However, the general RAG studies discussed above do not, by themselves, demonstrate the effectiveness of a particular EV assistant. The proposed component must be evaluated using EV-related questions and the actual knowledge sources incorporated into the prototype.

2.3.4 Synthesis of Related Research

The reviewed research establishes three important foundations for the proposed component. First, RAG can combine external knowledge retrieval with language generation. Second, LLMs can support natural-language responses but may produce unsupported or outdated information. Third, voice interfaces provide an alternative method of interaction, although their usability requires evaluation.

Together, these findings support investigating a virtual assistant that retrieves EV-specific information before generating conversational responses and provides voice interaction where available. They also indicate that evaluation should consider not only the naturalness of the responses but also their relevance to the retrieved information, understandability, response time, and usability.

The literature reviewed here provides the theoretical justification for the proposed design. A stronger final version of this section should also include verified studies specifically investigating EV information systems, automotive conversational assistants, or RAG-based applications in comparable technical domains.

2.4 Existing systems products and techniques

Existing conversational systems and information-retrieval techniques provide useful reference points for designing the proposed EV virtual assistant. They differ in their approaches to retrieving information, generating responses, and supporting voice-based interaction.

2.4.1 General-Purpose Conversational AI Systems

General-purpose conversational AI systems, including ChatGPT, are designed to respond to a wide range of natural-language questions. They can explain concepts, summarise information, and support interactive conversations. These capabilities make them relevant examples of LLM-based interaction.

However, a general-purpose conversational system is not necessarily configured with the specific, verified EV knowledge base required by this research. Responses may depend on the model's existing knowledge, the information supplied by the user, and any external retrieval capabilities available in the particular configuration.

The proposed EV virtual assistant differs in its intended focus: it uses a curated EV knowledge base as a source of context for answering domain-specific questions. Its objective is to combine natural-language response generation with retrieval from information selected for the EV domain.

2.4.2 Conventional Search and FAQ Systems

Conventional search systems allow users to locate information using keywords, while frequently asked questions (FAQ) systems provide predefined answers to common queries. These approaches can be effective when users know what to search for or when their questions closely match available content.

However, users may need to review several search results or navigate predefined categories to locate the information they require. FAQ systems may also have limited flexibility when users express questions in different ways or ask follow-up questions.

The proposed assistant aims to provide a more conversational method of accessing EV information. Instead of returning only matching documents or predefined answers, it retrieves relevant context and uses an LLM to formulate a natural-language response.

2.4.3 Retrieval-Augmented Generation Techniques

RAG combines information retrieval and language generation, allowing an LLM to use external information while answering questions. Lewis et al. (2020) established a foundational approach to retrieval-augmented generation, while Gao et al. (2024) reviewed subsequent RAG architectures and challenges.

Common techniques relevant to RAG-based systems include:

- Keyword-based retrieval: Identifies information by matching query terms with words in the knowledge source.
- Semantic retrieval: Uses representations of text, such as embeddings, to identify content with related meanings.
- Context-grounded generation: Supplies retrieved passages to an LLM so that its response can be informed by relevant source material.
- Prompt engineering: Structures instructions and retrieved context to guide the model toward clear, relevant responses.

These techniques are not interchangeable, and the final thesis should identify the exact methods used in the developed prototype. For example, semantic retrieval should only be claimed if embeddings or an equivalent semantic-search mechanism were implemented.

2.4.4 Voice-Enabled Conversational Techniques

Voice-enabled assistants use speech technologies to facilitate spoken interaction. Speech-to-text converts a user's spoken query into text, while text-to-speech converts a generated textual response into spoken output. Depending on the implementation, a virtual assistant may support voice input only, voice output only, or both.

Voice-user-interface research identifies usability and user experience as important aspects of evaluating these systems.

Therefore, integrating voice functionality into an EV assistant requires attention to speech recognition, response clarity, interaction errors, and user convenience.

In the proposed component, voice interaction is intended to complement the text-based question-answering workflow. The exact capabilities should be described according to the speech-processing tools and features implemented in the prototype.

2.4.5 Comparison of Existing Approaches

The following table presents a conceptual comparison of the approaches relevant to this research.

Approach	**Main advantage**	**Potential limitation**	**Relevance to this research**
General-purpose conversational AI	Flexible natural-language responses	May not be grounded in a dedicated EV knowledge source	Provides the conversational model concept
Keyword-based search	Direct access to matching information	Users may need to interpret several results	Provides a conventional retrieval baseline
FAQ-based systems	Simple and predictable answers to predefined questions	Limited flexibility for unfamiliar questions	Provides a basic question-answering approach

RAG-based question answering	Combines external knowledge retrieval with language generation	Depends on retrieval quality and source completeness	Forms the core approach of the proposed assistant
Voice-enabled assistants	Support spoken interaction	May be affected by recognition errors and usability issues	Provides the additional interaction capability

Table 2.1: Conceptual comparison of approaches relevant to the proposed EV virtual assistant.

This comparison describes general characteristics of the approaches rather than the results of a controlled experiment. It does not establish that the proposed system outperforms existing products. Such a conclusion would require a defined evaluation method and comparable test results.

2.5 Gap synthesis and conceptual framework

2.5.1 Synthesis of the Research Gap

The reviewed literature provides a foundation for combining external information retrieval, LLM-based response generation, and voice interaction. Foundational RAG research demonstrates the integration of retrieved knowledge with generative models, while subsequent reviews discuss challenges such as hallucination, outdated information, and retrieval quality. Research on voice interfaces further indicates the importance of evaluating usability and interaction effectiveness.

Nevertheless, the existence of these technologies does not automatically establish their suitability for every specialised application. For the proposed research, the practical challenge is to combine them into an EV-focused assistant that retrieves relevant information from a curated knowledge base and generates understandable responses to users' questions.

The preliminary research gap can therefore be stated as follows:

There is a need to design, implement, and evaluate an EV-focused conversational assistant that integrates retrieval from a dedicated EV knowledge base, LLM-based

response generation, and voice interaction within a single system. The research investigates how these capabilities can be combined to support access to EV-related information and evaluates the developed prototype according to its response relevance, understandability, usability, and response time.

This statement should be treated as the research gap addressed by the project, rather than a claim that no similar systems exist. A final claim of novelty requires a broader review of directly comparable EV assistants and related implementations.

2.5.2 Conceptual Framework

The conceptual framework illustrates how the main technologies and processes of the proposed virtual assistant work together to produce an answer to a user's question.



Figure 2.1: Conceptual framework of the proposed RAG-based EV virtual assistant. This is a conceptual representation; the final figure should be updated to reflect the implemented architecture.

The framework begins with a user submitting a question through text or voice input. If voice input is used, a speech-recognition mechanism converts the spoken question into text. The query is then processed by the retrieval component, which searches the

EV knowledge base for relevant information. The retrieved content is supplied to the LLM as context for generating an answer. The response is delivered through the conversational interface and may be converted into speech if voice output is supported.

The final stage involves evaluating the developed prototype. Response relevance and understandability help assess the usefulness of generated answers, while usability and response time provide additional measures of the interaction experience. These criteria establish a basis for assessing the component against its research objectives.

2.5.3 Summary

This chapter reviewed the theoretical foundations of conversational AI, LLMs, RAG, knowledge retrieval, and voice interaction. It also examined related research and compared general-purpose conversational systems, conventional search, FAQ-based systems, and RAG-based approaches. The synthesis identified the need to investigate the integration of EV-specific knowledge retrieval, context-aware response generation, and voice interaction in a dedicated virtual assistant. The conceptual framework derived from this review guides the methodology and system design presented in the following chapters.

3 RESEARCH METHODOLOGY

This chapter describes the research methodology adopted for the individual research component, “AI-Powered EV Virtual Assistant using Retrieval-Augmented Generation (RAG)”, which forms part of the broader project, “Electric Vehicle Management System Using Artificial Intelligence and Machine Learning.” It explains the research design, study materials, development procedure, evaluation methods, data analysis, ethical considerations, and research artefacts. The methodology focuses on developing a conversational assistant that retrieves relevant information from an electric vehicle (EV) knowledge base and uses a Large Language Model (LLM) to generate understandable responses to user queries, with voice interaction where implemented.

The following draft is structured for a thesis. Details such as the actual technologies, participant numbers, evaluation results, and data-handling procedures should be completed according to the work you carried out rather than assumed.

3.1 Research design

This research adopts a design-and-development approach to develop and evaluate an AI-powered virtual assistant for answering EV-related questions. The approach involves identifying the information-access problem, defining system requirements, designing the solution architecture, implementing the main components, and evaluating the resulting prototype.

The proposed system integrates Retrieval-Augmented Generation (RAG) with an LLM. The RAG component retrieves relevant information from a curated EV knowledge base, while the LLM uses the retrieved information and the user's question to generate a natural-language response. Voice interaction is included to support spoken queries and responses where these features are available in the implementation.

The research follows an iterative development process in which the assistant's functionality can be refined through testing, identification of errors, and improvements to the knowledge base, retrieval process, prompts, and user interface.

This approach is suitable for the research because the main outcome is a functional software component rather than a purely theoretical model.

The evaluation focuses on the assistant's ability to provide relevant and understandable answers, the usability of the interaction, and response time. These criteria are intended to assess whether the implemented prototype addresses the identified problem. The final evaluation methods should reflect the tests and feedback actually collected during the research.

3.2 Data participants or study materials

This research uses a curated collection of EV-related information as the primary knowledge source for the virtual assistant. The knowledge base is intended to provide relevant information for answering user questions about electric vehicle charging, battery care, vehicle maintenance, driving efficiency, and other common EV-related topics.

3.2.1 Study Materials

The main materials used in the research include:

- EV knowledge base: A collection of EV-related information used as the source for retrieving relevant content.
- User queries: Text-based or spoken questions used to examine the assistant's question-answering functionality.
- RAG pipeline: The retrieval process that identifies relevant knowledge and provides it to the LLM.
- Large Language Model: The language-generation component used to formulate responses based on the user query and retrieved context.
- Voice interaction tools: Speech-processing technologies used for voice input or output, depending on the implemented functionality.
- Testing records and feedback: Observations collected during prototype testing to assess response relevance, understandability, usability, and response time.

The knowledge base should be developed using relevant and reliable sources. The source documents, their origins, and any preprocessing applied should be documented to support the credibility and reproducibility of the research.

3.2.2 Participants

If the prototype was evaluated by users, the participants may include individuals who can interact with the assistant and provide feedback on its functionality and usability. Their feedback can help identify difficulties in asking questions, understanding responses, and using voice interaction.

The number of participants, their selection method, relevant characteristics, and recruitment procedure must be reported according to the actual study. If the evaluation involved only the researcher testing predefined questions, this section should describe the researcher-led testing procedure instead of claiming that external participants were involved.

3.3 Experimental or study procedure

The research procedure consists of several stages, beginning with the preparation of the knowledge base and continuing through system implementation and evaluation.

3.3.1 Requirement Identification

The initial stage involved identifying the purpose and intended functionality of the individual component. The main requirement was to provide users with a convenient way to obtain relevant EV information through natural-language questions. The system was therefore designed around knowledge retrieval, context-aware response generation, and voice interaction where supported.

3.3.2 Knowledge-Base Preparation

The next stage involved preparing EV-related information for use by the retrieval component. The knowledge base covers topics relevant to the intended functionality, including charging practices, battery care, maintenance, and driving efficiency.

The information is organised into a form suitable for retrieval. Depending on the implemented approach, preprocessing may include cleaning documents, removing

irrelevant content, dividing documents into smaller sections, and attaching metadata. Only preprocessing steps actually performed should be included in the final thesis.

The purpose of this stage is to make relevant information accessible to the retrieval pipeline when users submit questions.

3.3.3 RAG Pipeline Development

The RAG pipeline connects the user's query with relevant information from the EV knowledge base. When a question is submitted, the retrieval component searches the knowledge source and identifies content relevant to the question. The retrieved information is then prepared as context for the language model.

This process is intended to improve the relevance of generated responses by providing the model with domain-specific information. The quality of the resulting answer depends on whether the knowledge base contains suitable information and whether the retrieval component identifies the correct content.

The final thesis should specify the retrieval method, document-processing approach, embedding model, vector database, or other retrieval technologies used, where applicable.

3.3.4 LLM Integration and Prompt Engineering

After retrieving relevant information, the system supplies the question and retrieved context to the LLM. The model generates a natural-language response intended to address the user's question in a clear and understandable manner.

Prompt engineering is used to guide response generation. The prompts may specify the assistant's role, the use of retrieved context, the expected response style, and how the system should respond when sufficient information is unavailable.

The prompts should be designed to reduce unsupported answers and encourage the assistant to remain relevant to the user's question. However, prompt instructions alone cannot guarantee factual accuracy. The resulting responses must therefore be assessed during testing.

3.3.5 Voice Interaction

Voice interaction provides an additional means of communicating with the virtual assistant. Where voice input is implemented, the user's spoken question is converted into text before being processed by the RAG pipeline. The generated answer is then presented through the available interface.

If speech output is also implemented, the generated text is converted into speech and delivered to the user. Voice functionality should be evaluated separately where possible because speech-recognition errors and audio-related problems can affect the overall interaction.

3.3.6 Prototype Testing

The developed prototype is tested using questions covering the main topics represented in the EV knowledge base. Testing aims to determine whether the assistant retrieves relevant information, generates understandable responses, and provides a usable interaction experience.

The testing procedure may include the following activities:

1. Submit a predefined EV-related question.
2. Record the information retrieved by the system, where available.
3. Examine the generated response for relevance and factual consistency with the retrieved information.
4. Record the response time.
5. Repeat selected tests using different questions or alternative wording.
6. Test voice interaction if it is part of the implementation.
7. Record errors, limitations, and opportunities for improvement.

If user feedback was collected, it can be used alongside the technical testing results to assess the overall experience. The final thesis should report the actual test cases and procedures used.

3.4 Measures and evaluation metrics

The evaluation measures are selected to assess the functional and user-facing performance of the virtual assistant. The following metrics provide a proposed evaluation framework; only measures that were actually collected should be reported as experimental results.

Evaluation measure	Purpose	Possible evaluation method
Response relevance	Determines whether the answer addresses the user's question	Manual assessment using predefined criteria
Factual consistency	Examines whether the response is supported by the retrieved information	Compare the generated answer with the source context
Understandability	Assesses whether the answer is clear and easy to interpret	Reviewer assessment or user feedback
Response time	Measures the time required to generate an answer	Record elapsed time for each test query
Usability	Examines the ease of interacting with the assistant	User feedback or a structured usability questionnaire
Voice interaction performance	Examines whether spoken questions and responses function as intended	Test voice interactions and record recognition or delivery errors

Response relevance is important because a fluent answer is not necessarily an answer that addresses the question. Responses should be assessed against the intent of the query.

Factual consistency examines whether the generated response agrees with the retrieved source information. This measure is particularly relevant to a RAG-based system because retrieval is intended to ground the generated answer in the knowledge base.

Understandability and usability assess whether users can understand the response and interact with the assistant conveniently. If a questionnaire was used, its questions, rating scale, and scoring procedure should be documented.

Response time measures the time between submitting a query and receiving the response. The measurement points should be defined consistently, particularly when comparing text and voice interactions.

If a quantitative evaluation was performed, metrics should be calculated using the collected observations. If the evaluation was qualitative, the thesis should describe the observations and feedback without inventing numerical scores or claiming statistically significant improvements.

3.5 Data analysis

The collected data should be analysed according to the type of evidence available from prototype testing.

3.5.1 Quantitative Analysis

If numerical measurements were collected, descriptive statistics can be used to summarise the results. For example, response times can be reported using the mean, median, minimum, and maximum values. If responses were assessed using a rating scale, the distribution of ratings and relevant summary statistics can be reported.

For response time, the mean can be calculated as:

$$\bar{T} = \frac{1}{n} \sum_{i=1}^{n} T_i$$

where:

- $\bar{T}$ is the mean response time.
- T_i is the measured response time for the i-th query.
- n is the total number of queries tested.

This calculation should only be included as an evaluation result if response-time measurements are available.

3.5.2 Qualitative Analysis

Qualitative evidence may include observations of system behaviour, reviewer comments, and user feedback. These data can be grouped into themes such as response relevance, clarity, ease of use, usefulness of recommendations, and difficulties encountered during voice interaction.

The analysis should identify recurring observations and distinguish between positive feedback, reported problems, and areas requiring improvement. Where only informal feedback was collected, it should be described as preliminary qualitative evidence rather than as the result of a formal user study.

3.5.3 Interpretation of Findings

The results should be interpreted in relation to the research questions and objectives defined in Chapter 1. The analysis should examine whether the assistant retrieves relevant EV information, generates understandable responses, and provides the intended interaction features.

Any limitations identified during testing should also be discussed. For example, missing information in the knowledge base may result in incomplete responses, while speech-recognition errors may affect voice interactions.

The analysis should not claim that RAG improves accuracy compared with a general-purpose LLM unless a suitable comparative experiment was conducted. Similarly, claims about improved usability require evidence from the evaluation process.

3.6 Ethics privacy security and responsible AI

Ethical considerations are important when developing an AI-powered assistant that processes user queries and generates informational responses.

3.6.1 Privacy and Data Protection

The system should collect only the information necessary for its intended operation and evaluation. If user queries, feedback, or voice recordings are collected, the research should document how these data are handled, stored, and protected. Participants should be informed about the purpose of the evaluation and how their contributions will be used.

If external LLM or speech-processing services are used, their data-handling practices should be considered. The thesis should describe the actual privacy measures implemented rather than assuming that data are not retained by external services.

3.6.2 Security

Security considerations include protecting API credentials, restricting access to stored data, and preventing unauthorised access to system resources. Sensitive credentials should not be exposed in source code or public research artefacts.

The knowledge base should also be reviewed to reduce the risk of unintentionally including sensitive or unsuitable information. If the system uses external services, their authentication and access requirements should be handled appropriately.

3.6.3 Responsible AI and Reliability

LLM-generated responses may contain errors, unsupported statements, or incomplete information. The use of RAG can provide relevant source context, but it does not eliminate these risks.

The assistant should therefore be designed to encourage responses that remain relevant to the retrieved information. Where sufficient information is unavailable, the system should ideally indicate the limitation instead of presenting an unsupported answer as certain.

The assistant should be presented as an informational support tool rather than a replacement for professional EV maintenance or safety advice. Any recommendations concerning battery care, charging, or vehicle maintenance should be consistent with reliable information sources.

3.6.4 Research Integrity

All sources used to construct the knowledge base and support the thesis should be acknowledged appropriately. Evaluation results must be reported honestly, including unsuccessful tests and system limitations. User feedback should not be fabricated, and conclusions should be proportionate to the evidence collected.

Any required university ethical approval, participant consent, or data-management procedure should be reported according to the actual research process and applicable SLIIT requirements.

3.7 Reproducibility and research artefacts

Reproducibility is important for enabling other researchers to understand how the virtual assistant was designed, implemented, and evaluated. The research should document the main technical and procedural details needed to explain the system's behaviour.

The relevant research artefacts may include:

- Knowledge-base documentation: The topics covered, source references, document formats, and preprocessing steps.
- System architecture: A diagram showing the relationship between the user interface, retrieval component, knowledge base, LLM, and voice-processing components.
- Implementation details: The programming language, frameworks, libraries, model or service names, and relevant configuration details.
- Prompt templates: The instructions used to guide the LLM's response generation, excluding confidential credentials.
- Test cases: The questions used during evaluation, expected answer characteristics, observed responses, and recorded errors.
- Evaluation records: Response-time measurements, response assessments, questionnaire results, or qualitative feedback, depending on the evaluation performed.

- Version information: The versions of relevant libraries, models, and external services where available.
- Research documentation: Relevant design decisions, known limitations, and implementation changes made during development.

Where appropriate and permitted, non-sensitive source code, test cases, and supporting documentation can be maintained in a version-controlled repository. However, confidential credentials, private user data, and materials that cannot legally or ethically be shared should be excluded.

The final thesis should identify which artefacts are available and how they can be accessed. If the project cannot be reproduced exactly because of external API dependencies, model updates, unavailable data, or proprietary services, those limitations should be documented.

4 PRODUCT AND SOLUTION DESIGN

This chapter describes the requirements, architecture, design decisions, and detailed design of the individual research component, AI-Powered EV Virtual Assistant using Retrieval-Augmented Generation (RAG), developed as part of the overall project, Electric Vehicle Management System Using Artificial Intelligence and Machine Learning. The proposed component aims to provide users with an accessible way to obtain EV-related information through natural-language interaction. It combines a curated EV knowledge base, a retrieval process, and a Large Language Model (LLM) to generate context-aware responses, with voice interaction where supported by the implementation.

The chapter explains how the individual component fits within the complete system, the alternatives considered during design, the interactions between its main modules, and the standards and professional practices relevant to its development. The final design documentation should reflect the actual architecture, technologies, and features implemented in the prototype.

4.1 Requirements

Requirements define the functions and quality characteristics that the EV virtual assistant should provide. They establish the basis for designing the component, implementing its features, and evaluating whether the intended objectives have been achieved.

4.1.1 Functional Requirements

The main functional requirements of the proposed virtual assistant are presented below.

ID	Functional requirement	Description
FR01	Accept user queries	Allow users to submit EV-related questions through the supported interface.

FR02	Retrieve EV information	Search the curated knowledge base for information relevant to the user's question.
FR03	Generate responses	Use an LLM to generate natural-language responses informed by the retrieved context.
FR04	Provide EV guidance	Support questions about topics such as charging, battery care, maintenance, and driving efficiency, according to the available knowledge base.
FR05	Support voice interaction	Process spoken questions and provide spoken responses where the corresponding features have been implemented.
FR06	Manage conversational context	Maintain relevant conversational information during a session if session management is implemented.
FR07	Handle insufficient information	Where supported, indicate when the available knowledge is insufficient to answer a question reliably.
FR08	Support testing and evaluation	Allow the component to be tested using EV-related queries and the defined evaluation criteria.

4.1.2 Non-Functional Requirements

Non-functional requirements describe the expected quality and operational characteristics of the component.

- Accuracy and relevance: Responses should address the user's question and remain consistent with relevant retrieved information.
- Usability: The interface should make it easy for users to submit questions and understand the generated responses.
- Performance: The assistant should process queries and return responses within a reasonable time for the intended use.

- Reliability: The system should handle common query-processing errors and unavailable information appropriately.
- Maintainability: The knowledge base, prompts, and system components should be organised so that they can be updated when necessary.
- Security and privacy: Credentials and any user information processed by the system should be protected through appropriate controls.
- Accessibility: Voice interaction, where implemented, should provide an alternative way to communicate with the assistant.

These requirements represent design goals. Their achievement should be demonstrated through implementation evidence and evaluation results rather than assumed.

4.2 Component architecture within the complete solution

The AI-Powered EV Virtual Assistant is an individual component of the overall Electric Vehicle Management System. Its main responsibility is to provide conversational access to EV-related information. While the complete system may contain other EV management functions, this component focuses on knowledge retrieval, AI-based response generation, and voice interaction.

4.2.1 Overall System Architecture

The component can be represented through the following logical architecture.

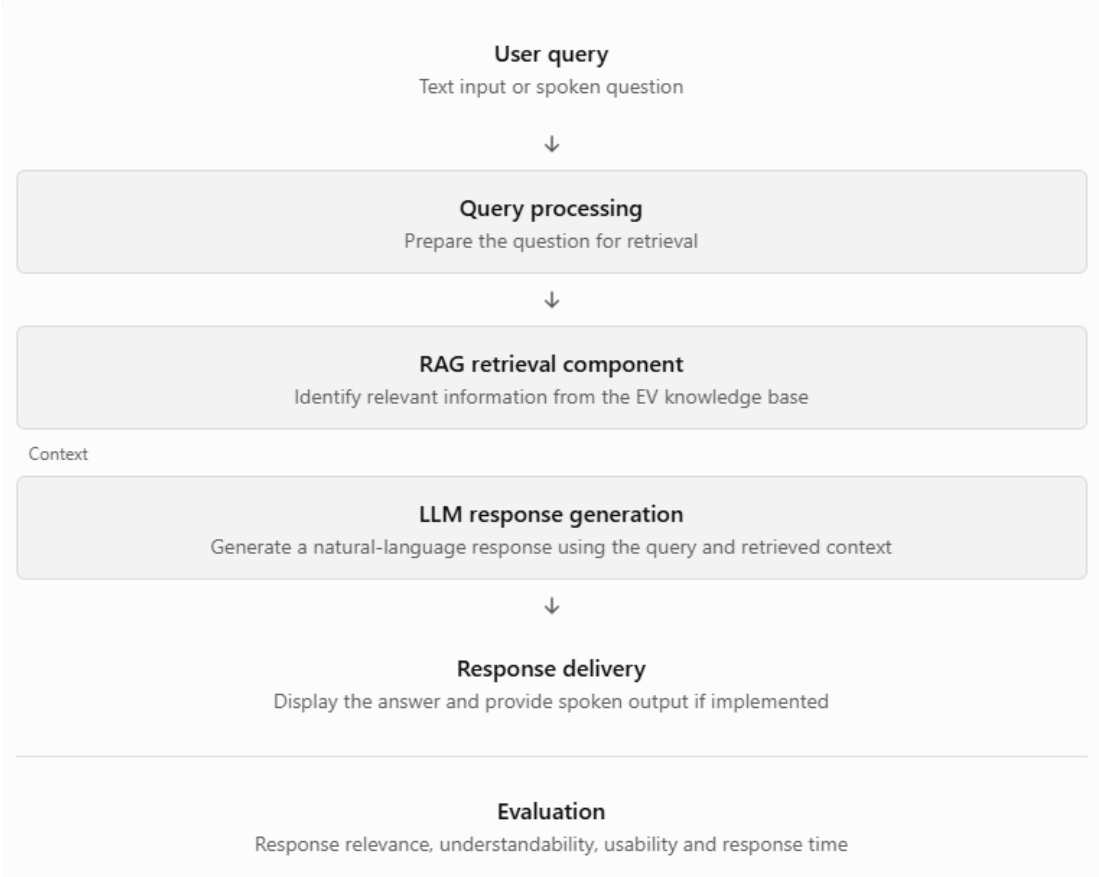


Figure 4.1: Logical architecture of the AI-Powered EV Virtual Assistant. Update the diagram to match the actual software architecture and data flow.

The architecture consists of several interconnected modules. The user interface receives questions, while the query-processing module prepares them for retrieval. The RAG module identifies relevant information from the EV knowledge base and supplies it as context to the LLM. The LLM then generates a response that is returned through the user interface.

If voice input is implemented, a speech-recognition component converts spoken queries into text before query processing. If voice output is supported, a text-to-speech component converts the generated response into speech. These voice-processing components complement the main question-answering workflow.

4.2.2 Integration With the Complete EV Management System

The virtual assistant is designed to contribute an intelligent conversational capability to the complete EV Management System. Its integration allows users to access EV-

related information through a question-and-answer interface rather than relying exclusively on conventional information browsing.

The integration should be considered at two levels:

- Functional integration: The virtual assistant contributes EV information retrieval and conversational response generation to the overall system.
- Technical integration: The assistant is connected to the main application's interface, backend, or other components through the interfaces supported by the implementation.

The distinction is important because a component can be designed to complement a larger system without being fully integrated into its running application. During the Progress Presentation 2 review, feedback identified integration with the main project as an area requiring improvement. Therefore, the final thesis should document the actual integration achieved, any interface changes made, and the remaining limitations.

The assistant should not be described as accessing live vehicle data, battery measurements, or charging-station services unless these connections were implemented and tested.

4.3 Design alternatives and decisions

The design of the virtual assistant involves selecting suitable approaches for information retrieval, response generation, and user interaction. The following alternatives provide a framework for documenting these decisions. The final thesis should distinguish between alternatives actually investigated and alternatives discussed only as possible approaches.

4.3.1 Information Retrieval Approach

Two possible approaches for accessing EV information are conventional keyword-based search and Retrieval-Augmented Generation.

Approach	**Advantages**	**Limitations**
Keyword-	Simple to implement and can	May miss relevant information

based search	identify direct word matches	when the question uses different wording
RAG-based retrieval	Supplies retrieved domain-specific context to an LLM for response generation	Depends on knowledge-base quality and retrieval relevance

RAG was selected as the central approach for this component because it combines information retrieval with natural-language response generation. This design allows the assistant to retrieve relevant EV information before formulating an answer. However, its effectiveness depends on the relevance of the retrieved context and the quality of the generated response.

4.3.2 Response Generation Approach

A second design decision concerns how answers should be generated.

A predefined FAQ system returns stored responses to known questions. Although this can provide predictable answers, it may be less flexible when users express questions differently or ask follow-up questions. An LLM-based approach can generate responses in natural language, while a RAG architecture provides retrieved information to support those responses.

The proposed component uses the RAG and LLM approach to support more flexible EV-related question answering. Prompt engineering can guide the model to use the retrieved information, respond clearly, and acknowledge insufficient information where this behaviour is implemented.

4.3.3 User Interaction Approach

The assistant can accept typed questions, spoken questions, or both. Text-only interaction is relatively straightforward to implement and test. Voice interaction offers an alternative input method but introduces additional requirements for speech recognition and, where applicable, speech generation.

Voice interaction was included in the scope of the individual component to support a more convenient conversational experience. The final implementation should specify

whether the prototype supports voice input, voice output, or both, and should describe the technologies used.

4.3.4 Design Decision Summary

The main design decisions are summarised below.

Design area	Selected approach	Reason
Information access	RAG-based retrieval	Connects user questions with information from an EV knowledge base
Response generation	LLM	Generates natural-language answers using the question and retrieved context
Response guidance	Prompt engineering	Helps guide the response format and use of retrieved information
Interaction	Text and voice, according to implementation	Supports conversational access through the available interface
Evaluation	Functional testing and response assessment	Examines whether the component meets its intended requirements

These decisions establish the conceptual basis of the component. Actual model names, libraries, databases, frameworks, and integration methods should be documented separately in the implementation chapter.

4.4 Detailed design

This section describes the internal behaviour of the virtual assistant and the responsibilities of its main modules.

4.4.1 Knowledge-Base Design

The EV knowledge base provides the information used by the retrieval component. It should cover the topics relevant to the assistant's intended purpose, including charging practices, battery care, maintenance, and driving efficiency.

The knowledge-base design should consider the following aspects:

- Content relevance: Information should relate to the intended EV-related questions.
- Source reliability: Information should come from credible, traceable sources.
- Content organisation: Documents or information passages should be organised for effective retrieval.
- Content maintenance: The knowledge base should be reviewed and updated when necessary.
- Source traceability: Where possible, the origin of information should be recorded so that retrieved content can be checked.

If documents are divided into smaller passages or converted into embeddings, the thesis should explain the chunking strategy, embedding model, and storage method actually used. These implementation choices should not be assumed from the use of RAG alone.

4.4.2 Query Processing and Retrieval Design

When a user submits a question, the system receives the query and prepares it for the retrieval process. The retrieval component then searches the knowledge base to identify content relevant to the question.

The retrieved information is passed to the response-generation module. The relevance of this information is important because the LLM may produce an incomplete or unsupported response if the retrieved context does not adequately address the user's question.

The retrieval design should therefore consider how questions are processed, how relevant information is selected, and how the retrieved passages are provided to the LLM. Where the system supports follow-up questions, the design should also explain how conversational context is maintained.

4.4.3 LLM Response-Generation Design

The response-generation module uses the user's question and retrieved information to produce a natural-language answer.

The prompt design may define the assistant's role, the expected response style, and the relationship between the retrieved context and the generated answer. For example, the assistant can be instructed to prioritise the supplied EV information and avoid presenting unsupported claims as established facts.

The generated response is then returned to the user through the interface. If the retrieved information is insufficient, the system should ideally communicate that limitation rather than generate an unnecessarily confident answer. Whether this behaviour is enforced in the implementation should be documented and tested.

4.4.4 Voice Interaction Design

Voice interaction extends the question-answering workflow by allowing users to speak their questions. The design can include the following sequence:

1. The user submits a spoken question.
2. A speech-recognition service converts the speech into text.
3. The recognised text is passed to the query-processing module.
4. The RAG pipeline retrieves relevant EV information.
5. The LLM generates a response.
6. The response is displayed and, if supported, converted into speech.

The voice interface introduces possible errors during speech recognition and audio delivery. These errors should be considered during testing. If the prototype implements only voice input or only voice output, the final description should be adjusted accordingly.

4.4.5 Conversational Session Management

Session management can be used to maintain relevant information from earlier interactions during a conversation. This can help the assistant interpret follow-up questions that refer to previous responses.

For example, a user might ask about EV charging and then follow up with a question about the charging practice discussed earlier. Maintaining relevant conversational context can help the assistant interpret the second question.

However, conversational memory must be managed carefully to prevent unrelated or outdated context from influencing new responses. If session management was implemented, the thesis should document what information is retained, how it is used, and when it is cleared. If this feature was not implemented, it should be presented as a potential enhancement rather than a completed function.

4.4.6 End-to-End Interaction Flow

The complete interaction begins when a user submits an EV-related question. The system prepares the query, retrieves relevant information, and supplies the retrieved context to the LLM. The LLM generates an answer, which is then delivered through the interface. If voice features are enabled, speech processing is included at the relevant input or output stage.

The following example illustrates the intended interaction.

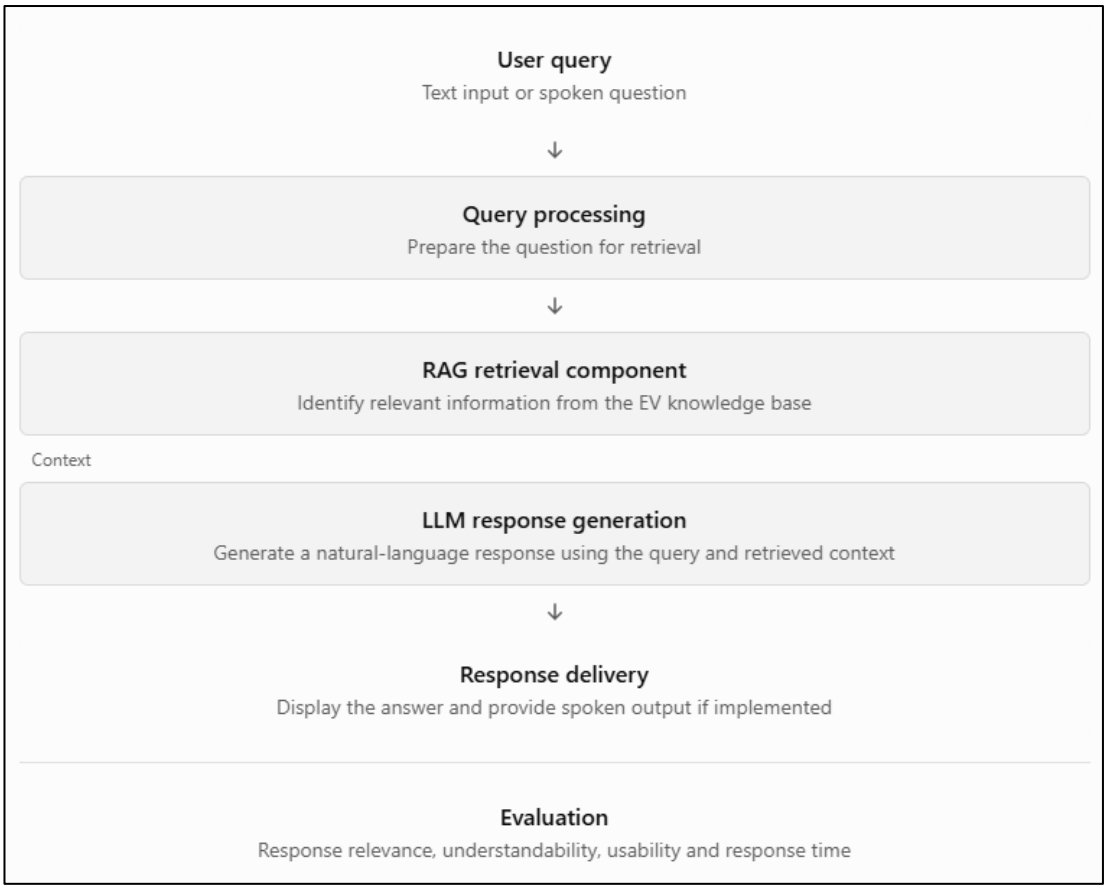


Figure 4.2: Example interaction flow of the proposed virtual assistant. This illustrates the intended workflow and is not evidence of a particular test result.

4.5 Standards and professional practice

The development of the AI-powered EV virtual assistant should follow appropriate software engineering, information management, security, and responsible AI practices. These practices support the reliability, maintainability, and ethical use of the system.

4.5.1 Software Engineering Practices

The component should be developed using a structured approach that includes requirements identification, modular design, implementation, testing, and documentation. Separating the user interface, retrieval pipeline, knowledge base, and response-generation logic can make the system easier to maintain and improve.

Version control and clear documentation should be used where available to track code changes, record implementation decisions, and support collaboration with the research team.

4.5.2 Information Quality and Traceability

The EV knowledge base should use reliable information sources, and its content should be reviewed for relevance and accuracy. Source details should be retained wherever possible so that information used in generated answers can be checked against its origin.

The system should not be assumed to provide fully accurate responses merely because RAG is used. Retrieved content may be incomplete, and the LLM may misinterpret or incorrectly summarise it. Testing should therefore examine the relationship between retrieved information and generated answers.

4.5.3 Security and Privacy Practices

API keys, passwords, and other credentials should be protected and excluded from public repositories. Any collected user queries, feedback, or voice recordings should be handled according to the research's privacy requirements and applicable university policies.

Where external AI or speech-processing services are used, their data-handling requirements should be considered. Access controls and appropriate data-management procedures should be applied according to the system's actual architecture.

4.5.4 Responsible AI Practices

Responsible AI development requires attention to accuracy, transparency, limitations, and potential user reliance on generated information. The assistant should be designed to provide useful responses without implying that every answer is guaranteed to be correct.

In particular, EV-related information concerning battery care, charging, and maintenance should be consistent with reliable sources. The assistant should communicate uncertainty or insufficient information where feasible, and it should not

be presented as a substitute for professional vehicle diagnostics or safety-critical advice.

4.5.5 Academic and Professional Integrity

The research should acknowledge the sources used in the knowledge base, literature review, and system design. Any external libraries, models, APIs, and third-party tools should be identified in accordance with their documentation and licensing requirements.

The implementation, testing process, and results should be reported honestly. Limitations identified during development and feedback received during project presentations should be documented where relevant. The final thesis should clearly distinguish implemented features from proposed enhancements.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The AI-Powered EV Virtual Assistant was implemented using Python and a set of libraries and external services that support document processing, semantic retrieval, LLM-based response generation, and voice interaction. The implementation follows a Retrieval-Augmented Generation (RAG) architecture in which EV-related PDF documents are processed and indexed, relevant information is retrieved in response to user questions, and a large language model generates conversational answers using the retrieved context.

The development environment uses Python-based libraries for application logic and integrates external services for embedding generation, vector storage, and language generation. The voice engine supports speech recognition and speech synthesis, while the interactive interface provides text-based and voice-enabled operating modes.

The principal technologies identified in the implementation are summarised below.

Technology	**Purpose in the implementation**
Python	Implements the assistant, session management, voice engine, and interactive interface
LangChain 0.2.17	Supports document processing, prompt templates, conversation memory, and LLM integration
LangChain-Groq 0.1.10	Integrates the Groq-hosted language model
Groq API	Provides access to the llama-3.3-70b-versatile language model
Pinecone	Stores and retrieves vector representations of EV document passages
Pinecone Inference API	Generates embeddings using llama-text-embed-

	v2
PyPDF / LangChain document loaders	Loads PDF documents from the configured document directory
RecursiveCharacterTextSplitter	Divides documents into smaller text chunks
Requests	Sends HTTP requests to the embedding and vector database APIs
SpeechRecognition	Supports speech recognition
PyAudio	Provides microphone access when available
gTTS	Generates spoken audio from text
pygame / pydub	Supports audio playback
pyttsx3	Provides an alternative text-to-speech engine
FFmpeg and eSpeak	Provide audio-processing and speech-related system dependencies

Table 5.1: Technologies used in the EV virtual assistant implementation.

The installation code pins selected package versions to reduce dependency conflicts. The voice engine also checks whether microphone and audio playback capabilities are available and selects an appropriate text-to-speech backend.

The application depends on external services for embedding generation, vector retrieval, and LLM responses. Consequently, successful operation requires valid credentials, network connectivity, correctly configured services, and an accessible local PDF document directory.

5.2 Component implementation

5.2.1 EV Knowledge-Base Preparation

The knowledge base is created from PDF documents stored in a configured directory. The implementation uses LangChain's DirectoryLoader with PyPDFLoader to load PDF files and extract their textual content.

The extracted documents are divided into smaller passages using RecursiveCharacterTextSplitter. The configured chunk size is 1,000 characters, with an overlap of 200 characters. This overlap helps preserve some contextual continuity between neighbouring passages.

The resulting chunks are converted into vector embeddings using Pinecone's Inference API and the llama-text-embed-v2 embedding model. The embedding request uses the passage input type for document text and applies the configured truncation behaviour.

Each processed chunk is assigned a hash-based identifier and stored in Pinecone with its embedding and metadata. The metadata contains the text passage, source document path, and page number.

The implementation checks the vector index before indexing documents. If the index already contains vectors, it skips the indexing process. This reduces unnecessary processing when the index is already populated. However, it also means that changes to the source PDFs are not automatically reflected in the index unless an appropriate re-indexing procedure is performed.

5.2.2 Semantic Retrieval

When a user submits a question, the assistant generates an embedding for the query using the same embedding model. The query embedding is sent to Pinecone, which retrieves the five most similar stored vectors using the configured topK = 5 parameter.

The retrieved matches include metadata containing the original text passages and their source information. These passages are combined into a context string and passed to the language model.

This approach enables the system to retrieve information based on semantic similarity rather than relying exclusively on exact keyword matches. The quality of the retrieval depends on the contents of the knowledge base, the embedding model, and the relevance of the indexed passages to the user's question.

5.2.3 LLM Integration and Response Generation

The implementation integrates a Groq-hosted language model through the ChatGroq class provided by LangChain-Groq. The configured model is llama-3.3-70b-versatile, with a temperature of 0.3 and a maximum output length of 1,024 tokens.

A custom prompt is constructed using the user's question, the retrieved context, and the conversation history. The prompt defines the assistant as an EV-focused conversational assistant and instructs it to provide clear, professional responses.

The prompt also specifies how the assistant should respond to questions unrelated to electric vehicles and questions for which it lacks sufficient information. These instructions are intended to keep responses within the system's defined subject area and reduce unsupported answers.

The model receives the formatted prompt and generates a response, which is returned to the user through the chat interface. The implementation also returns metadata for the retrieved source passages, allowing the application to retain information about which documents contributed context to the answer.

Implementation limitation: The code includes source metadata in the response object, but the interactive interface does not currently display those sources to the user. In addition, the off-topic and insufficient-information rules are implemented through prompt instructions rather than a separate validated classification or retrieval-confidence mechanism. Therefore, these behaviours require testing and should not be described as guaranteed.

5.2.4 Conversational Session Management

The assistant uses LangChain's ConversationBufferMemory to maintain conversation history for individual sessions. A session is assigned a unique identifier generated using Python's uuid module.

Each session stores its conversation memory, creation time, message count, and most recent interaction time. When a new question is submitted, the assistant retrieves the conversation history and includes it in the prompt supplied to the LLM.

The SessionManager class maps user identifiers to session identifiers. It creates sessions when necessary, supports conversations associated with individual user identifiers, and provides functions for ending sessions and clearing stored conversational memory.

This implementation supports contextual follow-up questions within an active session. However, session information is maintained in application memory rather than a persistent database. Sessions may therefore be lost when the application process terminates.

5.2.5 Voice Interaction

The voice functionality is implemented through a custom VoiceEngine class that combines speech recognition and text-to-speech capabilities.

The SpeechRecognition library is used to capture and process spoken questions. When a microphone is available, PyAudio provides microphone access, and the engine uses Google's speech-recognition service through recognize_google() to convert speech into text.

For speech output, the implementation prioritises gTTS when an audio playback backend is available. It uses pygame or pydub to play generated audio and falls back to pyttsx3 when appropriate. The engine also includes functionality for transcribing audio files and saving generated speech to an audio file.

The interactive interface supports three main interaction configurations:

- Text input with text output.
- Text input with voice output.
- Voice input with voice output.

The implementation checks whether the required voice capabilities are available and falls back to text input when a live microphone is unavailable. This allows the

assistant's core question-answering functionality to remain accessible in environments where microphone access is not supported.

The speech-processing functions depend on external services and local audio configuration. In particular, Google speech recognition and gTTS require network access, while microphone input and audio playback depend on the operating environment.

5.2.6 Interactive Chat Interface

The EVChatInterface class provides a command-line interface for interacting with the assistant. It supports submitting questions, viewing conversation history, starting a new session, enabling or disabling voice input, and ending the conversation.

The application also provides a menu-driven entry point through the main() function. Users can select the preferred interaction mode, run automated test queries, view active sessions, or exit the application.

The automated testing function contains eight predefined questions covering EV fundamentals, battery technology, charging levels, vehicle range, maintenance costs, regenerative braking, an unrelated oil-price question, and a question about long-range EV models.

These predefined queries provide a starting point for functional testing. Their presence in the code demonstrates that a test procedure has been prepared; it does not, by itself, demonstrate that all queries were executed successfully or that the responses were accurate.

5.3 Integration with the other components

The virtual assistant integrates several software components within a single Python application. The main integration takes place between the document-processing pipeline, Pinecone vector index, Groq language model, conversation-management module, voice engine, and interactive interface.

The document-processing module prepares the EV knowledge base and stores the resulting vectors in Pinecone. During a conversation, the query embedding is generated through the Pinecone Inference API, and the relevant passages are

retrieved from the vector index. These passages are then supplied to the Groq-hosted LLM, which generates the response.

The session manager connects the interactive interface with the assistant's conversational memory. The voice engine connects spoken input and audio output with the same question-answering workflow used for text queries.

The integration flow can be summarised as follows:



Figure 5.1: Integration architecture of the implemented EV knowledge assistant. The diagram represents the software flow evident in the supplied code.

Although the code demonstrates integration among the assistant's internal modules and external services, it does not show an API or interface connection to the wider Electric Vehicle Management System. Therefore, integration with the main group project should be documented separately based on the actual integration achieved. If the component was still operating independently, this should be acknowledged as a limitation rather than presented as completed system-level integration.

5.4 Verification and software testing

The implementation includes an automated test function that submits predefined questions to the assistant. This provides a practical starting point for verifying the retrieval and response-generation workflow.

5.4.1 Functional Test Cases

The following table organises the test scenarios based on the supplied implementation.

Test ID	Test scenario	Expected behaviour
TC01	Ask “What is an electric vehicle?”	Retrieve relevant EV information and generate an answer
TC02	Ask about lithium-ion batteries in EVs	Retrieve relevant battery information and generate an answer
TC03	Ask about EV charging levels	Return an answer relevant to charging
TC04	Ask about average EV range in 2024	Generate a response consistent with the available information
TC05	Ask about EV and gasoline-car maintenance costs	Retrieve relevant information and generate a comparative answer
TC06	Ask about regenerative braking	Retrieve relevant information and explain the concept
TC07	Ask for today's crude oil price	Apply the off-topic response instruction
TC08	Ask which EV currently has the longest range	Respond according to the retrieved information and its currency

Table 5.2: Functional test scenarios defined in the implementation.

These are test scenarios extracted from the supplied code. Their expected behaviours must be verified by executing the tests and inspecting the actual outputs.

TC04 and TC08 require particular care because EV range information changes over time. The code does not show a live data source for current vehicle specifications, so the assistant should not be assumed to provide up-to-date market information.

5.4.2 Retrieval and Response Verification

Retrieval verification should examine whether the five passages returned by Pinecone contain information relevant to each test question. Response verification should then examine whether the LLM's answer accurately represents that information.

For each test, the following information can be recorded:

- The question submitted.
- The retrieved passages and their source documents.
- Whether the retrieved passages are relevant.
- The generated answer.
- Whether the answer is factually consistent with the retrieved passages.
- Any unsupported statements or missing information.

The current implementation returns source information in the response object, which can be used for this analysis. However, a structured evaluation procedure is required before drawing conclusions about the assistant's retrieval quality or answer accuracy.

5.4.3 Session and Voice Testing

Session testing should verify whether follow-up questions use relevant conversation history and whether ending a session clears its stored memory. The code includes functions for creating sessions, viewing history, ending sessions, and clearing active sessions.

Voice testing should examine whether the microphone captures spoken questions, whether speech recognition returns appropriate text, and whether the generated answer can be played through the selected text-to-speech backend.

The implementation includes fallback behaviour when microphone or audio capabilities are unavailable. This behaviour should also be tested in the target environment.

5.4.4 Recording Test Results

The following template can be completed after executing the automated tests.

Test ID	Actual result	Status	Observations
TC01	To be recorded	Pending	Record answer relevance
TC02	To be recorded	Pending	Check battery information
TC03	To be recorded	Pending	Check charging information
TC04	To be recorded	Pending	Check source and date relevance
TC05	To be recorded	Pending	Check support for cost comparisons
TC06	To be recorded	Pending	Check technical explanation
TC07	To be recorded	Pending	Check off-topic handling
TC08	To be recorded	Pending	Check whether the information is current

Table 5.3: Template for recording actual test results.

The supplied code does not include execution logs or measured evaluation results. Consequently, pass rates, response accuracy, average response time, and usability scores cannot yet be reported. These values should be added only after the corresponding tests have been performed.

5.5 Deployment and operation

The assistant is configured to run as a Python application in a local development environment. The code contains a Windows-style path for the EV PDF documents and provides a command-line menu through which users can select the available interaction modes.

Before starting the application, the required Python packages must be installed, the PDF directory must be accessible, and the Groq and Pinecone services must be correctly configured. The Pinecone index must also be available with a compatible embedding dimension and suitable metadata configuration.

During initialisation, the application checks the Pinecone index and indexes the PDF documents if the index contains no vectors. It then initialises the Groq language model and the voice engine. Once the components are ready, the user can interact with the assistant through the main menu.

The application supports text-only conversations, text conversations with spoken responses, and voice-based conversations when the required hardware and libraries are available. It also supports predefined testing and session inspection.

The implementation depends on external APIs and therefore requires network connectivity. Voice input requires a working microphone and speech-recognition access, while audio output requires a supported playback mechanism or an alternative text-to-speech backend.

For a more reliable deployment, the application should use environment variables or a secrets manager for API credentials, include clear error handling for unavailable services, and document the process for updating the knowledge base. Since the supplied code stores session memory in the running process, active conversations are not persistent across application restarts.

6 RESULTS AND VALIDATION

6.1 Research results

The research focused on developing an AI-powered Electric Vehicle (EV) Virtual Assistant using Retrieval-Augmented Generation (RAG) to provide context-aware information about electric vehicles. The developed prototype combines a curated EV knowledge base, semantic information retrieval, a Large Language Model (LLM), and voice interaction capabilities.

The implementation demonstrates the technical approach of retrieving relevant information from PDF-based EV documents and providing it as context to an LLM before generating a response. The knowledge base is processed by splitting documents into smaller text chunks, generating embeddings, and storing the resulting vectors in Pinecone. When a user submits a question, the system generates a query embedding and retrieves the five most relevant matches from the vector database. The retrieved information is then incorporated into a prompt for the LLM to generate a response.

The prototype also includes conversational session management, allowing the assistant to use previous messages as conversation history. Voice interaction is supported through speech recognition and text-to-speech functionality, providing options for text-based and voice-based communication.

The implementation addresses the main research objectives by demonstrating how RAG and conversational AI can be applied to EV-related information retrieval. However, the effectiveness of the generated responses must be established through systematic testing. Since quantitative test results have not yet been documented, claims about response accuracy, performance improvements, or superiority over other approaches cannot be made at this stage.

6.2 Product and technical results

The primary technical outcome of the research is a command-line prototype of an AI-powered EV virtual assistant. The implemented components and their intended contributions are summarized in Table 6.1.

Table 6.1: Product and Technical Outcomes

Component	Implementation and outcome
EV knowledge base	Supports loading EV-related PDF documents from a specified directory.
Document processing	Splits document content into chunks of 1,000 characters with a 200-character overlap.
Embedding generation	Uses the Pinecone Inference API with the llama-text-embed-v2 embedding model.
Vector storage and retrieval	Uses Pinecone to store document embeddings and retrieve the five most relevant matches for a query.
Response generation	Uses the Groq API with the llama-3.3-70b-versatile model to generate responses using retrieved context.
Prompt engineering	Defines EV-related response behaviour, conversational context, and instructions for handling off-topic or insufficiently supported questions.
Session management	Maintains conversation history and session information during the running application.
Voice interaction	Supports speech-to-text input and text-to-speech output, subject to microphone, audio-library, and network availability.
User interface	Provides a command-line interface with text interaction, voice options, session controls, and predefined test questions.

The technical design combines information retrieval with language generation to ground responses in the available EV documents. The prompt also instructs the assistant to avoid answering unrelated questions and to indicate when the available context is insufficient. These safeguards are implemented through prompt instructions; their effectiveness requires separate validation.

The prototype has several technical dependencies. It requires valid API credentials and network access for the external embedding and language-model services. Voice interaction additionally depends on compatible audio hardware and software. Furthermore, conversation history is maintained in application memory rather than a persistent database, meaning that session information is not guaranteed to remain available after the application is restarted.

Although the implementation provides the core functions of an EV virtual assistant, the presence of these functions alone does not establish their accuracy, reliability, or usability. Those characteristics must be evaluated using documented test cases and observed results.

6.3 User or stakeholder validation

User or stakeholder validation is important for determining whether the virtual assistant provides understandable, relevant, and useful information to its intended users. The proposed validation criteria include response relevance, factual consistency with the retrieved EV information, understandability, response time, and the usability of text and voice interaction.

The system includes eight predefined test questions covering EV fundamentals, lithium-ion batteries, charging levels, vehicle range, maintenance costs, regenerative braking, an unrelated question about crude oil prices, and a question about the EV with the longest available range. These questions provide a basis for examining the assistant's behaviour across different types of queries. However, the implementation of these test questions does not, by itself, demonstrate that the tests have been executed successfully.

Table 6.2 provides a framework for recording the validation evidence.

Table 6.2: User and Technical Validation Framework

Validation criterion	**Evaluation approach**	**Evidence to record**
Response	Compare the generated response	Relevant, partially relevant,

relevance	with the user's question.	or irrelevant
Factual consistency	Compare factual statements with the retrieved documents.	Supported, partially supported, or unsupported
Context retrieval	Examine whether the retrieved document passages address the query.	Relevant or irrelevant retrieved passages
Understandability	Assess whether the response is clear and understandable.	Reviewer comments or user feedback
Response time	Measure the elapsed time between submitting a query and receiving a response.	Response time in seconds
Off-topic handling	Test a question unrelated to EVs.	Whether the assistant follows the intended refusal behaviour
Voice interaction	Test speech recognition and spoken response generation.	Successful interaction, recognition errors, or playback errors
Usability	Collect feedback on the interaction process, where participants are available.	Actual user comments or questionnaire results

At the time of writing, no numerical evaluation results, participant counts, completed questionnaires, or documented stakeholder feedback have been supplied for this chapter. Therefore, the effectiveness of the system against these criteria remains to be established from actual evaluation evidence. If user testing or supervisor feedback has been conducted, the relevant findings should be added here, together with the evaluation method and the number of participants, where applicable.

6.4 Integration results

The prototype integrates several technical components within a single application workflow. The command-line interface accepts a question, the session manager maintains the conversation context, the embedding service converts the question into a vector representation, and Pinecone retrieves relevant document passages. The retrieved context and conversation history are then passed to the Groq-hosted LLM, which generates the response returned to the user. Where enabled, the voice engine provides speech recognition and spoken output.

This internal workflow demonstrates the intended interaction between the retrieval, generation, session-management, and voice components. Its successful operation must nevertheless be confirmed through execution and integration testing.

Table 6.3: Integration Status

Integration area	**Status based on the implementation**
Command-line interface and session manager	Implemented in the prototype code
Session manager and RAG assistant	Implemented in the prototype code
RAG assistant and Pinecone	API-based integration implemented; operational validation required
RAG assistant and Groq LLM	API-based integration implemented; operational validation required
Voice input and assistant workflow	Implemented in the prototype code; requires environment-specific testing
Assistant response and voice output	Implemented in the prototype code; requires playback testing

Integration with the overall EV Management System	Full integration is not demonstrated by the available implementation

The integration of the individual modules represents progress towards the overall research objective. However, feedback received during the second project presentation indicated that the individual component had not yet been adequately integrated into the overall EV Management System and that changes to the frontend were required. The current implementation is a command-line prototype, and the available code does not demonstrate a completed connection to the group's main application frontend.

Consequently, internal integration and integration with the complete group system must be treated as separate outcomes. The internal workflow is represented in the implementation, whereas full system integration remains an outstanding limitation unless subsequent work has addressed it. Any later frontend integration should be documented with evidence such as screenshots, API interaction records, or successful end-to-end test cases.

6.5 Summary of findings

This chapter presented the technical outcomes and validation requirements of the AI-powered EV Virtual Assistant using RAG. The prototype incorporates PDF-based knowledge preparation, embedding generation, semantic retrieval through Pinecone, context-aware response generation using an LLM, conversational session management, and voice interaction capabilities.

The implementation provides a foundation for answering EV-related questions using information retrieved from a curated knowledge base. It also includes prompt instructions intended to restrict responses to EV-related topics and handle questions for which sufficient context is unavailable. However, the effectiveness of these behaviours has not been established through documented evaluation results.

The findings also highlight limitations relating to external API dependencies, voice-environment requirements, non-persistent session memory, and the lack of demonstrated integration with the complete EV Management System. Therefore, the research has produced a technical prototype and an implementation workflow, while

further empirical validation and main-system integration are necessary to establish its overall effectiveness and readiness for practical use.

7 DISCUSSION

7.1 Interpretation of findings

The research aimed to develop an AI-powered Electric Vehicle (EV) Virtual Assistant using Retrieval-Augmented Generation (RAG) to provide relevant information about electric vehicles through text-based and voice-based interaction. The implementation combines a document-based knowledge base, semantic retrieval, a Large Language Model (LLM), conversational session management, and voice processing. These components provide a technical foundation for an EV-focused conversational assistant.

The main finding from the development process is that RAG provides a practical architecture for combining external EV information with language-model response generation. Instead of relying exclusively on information stored in the LLM's parameters, the system retrieves relevant passages from a collection of EV documents and incorporates them into the prompt used to generate an answer. This approach is consistent with the principles described by Lewis et al. (2020), who introduced RAG as a method for combining parametric language-model knowledge with retrieved external information.

The implementation also demonstrates the importance of document preparation and semantic retrieval. The quality of the final response depends not only on the language model but also on the relevance and quality of the source documents, the document-chunking strategy, and the information retrieved for each question. If relevant information is absent from the knowledge base or is not retrieved, the generated answer may be incomplete or unsupported.

Conversational session management provides another important capability by allowing previous messages to be included in subsequent interactions. This supports follow-up questions and more context-aware conversations. However, the implementation maintains session information in application memory, so it does not provide persistent conversation storage across application restarts.

The voice component provides an additional interaction method through speech recognition and text-to-speech output. Nevertheless, voice functionality introduces dependencies on audio hardware, software compatibility, network availability, and speech-recognition performance. Therefore, its practical value should be determined through testing under realistic usage conditions.

Overall, the implementation demonstrates the feasibility of combining RAG, an LLM, conversational context, and voice interaction within an EV-focused prototype. However, the degree to which these capabilities improve response accuracy, user satisfaction, or efficiency remains unconfirmed until the proposed evaluation has been completed and its results documented.

7.2 Comparison with literature baselines and existing solutions

Existing approaches to information access include conventional search engines, static FAQ systems, general-purpose AI assistants, and RAG-based conversational systems. These approaches differ in how they retrieve information, generate responses, maintain conversational context, and support user interaction.

Lewis et al. (2020) introduced RAG as a framework that combines a language model with a retrievable external knowledge source. Their research demonstrated the potential of this architecture for knowledge-intensive natural language processing tasks. The present research applies the same general principle to EV information by retrieving passages from a curated document collection and supplying them to an LLM to generate conversational responses.

Compared with a conventional FAQ system, the proposed assistant is designed to accept natural-language questions rather than requiring users to select a predefined question. It can also use retrieved document passages and conversation history when generating responses. However, these architectural differences do not establish that the prototype is more accurate or efficient than an FAQ system; such a conclusion would require comparative testing.

Compared with general-purpose AI assistants, the proposed system is designed around a domain-specific EV knowledge base. This provides a mechanism for grounding responses in selected EV documents. Nevertheless, the reliability of this

approach depends on the completeness, relevance, and quality of the knowledge base. A general-purpose assistant may have access to broader information, while the proposed system may be limited to the documents and retrieval mechanisms configured for it.

Voice interaction is another distinguishing feature of the prototype. Deshmukh and Chalmeta (2024) reviewed research on voice-user-interface experience and usability, highlighting the importance of evaluating interaction quality and user experience. In the present research, speech recognition and text-to-speech capabilities provide the technical basis for voice interaction, but their usability and effectiveness have not yet been established through documented user testing.

Table 7.1: Comparison of Information-Access Approaches

Approach	**Main characteristics**	**Comparison with the proposed system**
Conventional search engine	Returns links or documents relevant to a query.	The proposed assistant is designed to generate a conversational response from retrieved information.
Static FAQ system	Provides predefined answers to known questions.	The proposed assistant supports free-form questions and context-aware response generation.
General-purpose AI assistant	Generates responses across multiple subject areas.	The proposed system is configured to focus on EV-related information from a selected knowledge base.
RAG-based conversational assistant	Combines external retrieval with LLM-based response generation.	The proposed system follows this architecture using Pinecone, an embedding model, and a Groq-hosted LLM.

Proposed EV Virtual Assistant	Combines EV document retrieval, conversational context, and optional voice interaction.	Provides an EV-focused prototype, with effectiveness and comparative performance still requiring validation.

The comparison is based on the general characteristics of these approaches and the implemented design of the prototype. It is not the result of a controlled benchmark. No comparative measurements have been provided for response accuracy, response time, user satisfaction, or resource consumption. Consequently, the research should be interpreted as an implementation and application of established RAG principles rather than evidence that the proposed system outperforms existing solutions.

7.3 Research contribution

The research contributes to the application of conversational AI and RAG within the domain of electric vehicle information access. Its main contribution is the design and implementation of an EV-focused virtual assistant that combines semantic document retrieval, LLM-based response generation, conversational context, and voice interaction in a single prototype.

The individual research contribution can be described through the following areas.

Application of RAG to EV information. The system uses a curated collection of EV-related PDF documents as an external knowledge source. Document content is divided into overlapping chunks, embedded into vector representations, and stored in Pinecone. Relevant passages are retrieved when a user submits a question and supplied to the LLM as context for response generation.

Context-aware conversational response generation. The implementation incorporates conversation history into the response-generation process. This supports the intended handling of follow-up questions and enables the assistant to consider earlier messages during a session.

Domain-focused prompt engineering. The prompt instructs the assistant to answer EV-related questions, respond appropriately when the retrieved context is insufficient, and avoid unrelated topics. These instructions provide an initial

mechanism for controlling the scope and style of responses, although their reliability must be assessed through testing.

Combination of text and voice interaction. The prototype integrates speech recognition and text-to-speech functionality with the conversational workflow. This provides alternative interaction modes for users who prefer speaking rather than typing, subject to the capabilities of the execution environment.

Evaluation framework for a domain-specific assistant. The research identifies relevant evaluation dimensions, including response relevance, factual consistency, retrieval quality, understandability, response time, and voice-interaction performance. These criteria provide a basis for systematic validation of the prototype.

The contribution should be understood as an applied system-development contribution. RAG, vector retrieval, LLMs, and voice technologies are established techniques; therefore, the research does not claim to have introduced a new RAG algorithm or a new language model. Its contribution lies in combining these techniques for an EV-focused use case and identifying the technical and evaluation requirements associated with that application.

7.4 Product contribution

The product contribution is a command-line prototype that demonstrates the main workflow of an AI-powered EV virtual assistant. It provides a foundation for exploring how users could obtain information about electric vehicles through conversational questions.

The prototype supports loading EV-related PDF documents, preparing document chunks, generating embeddings, retrieving relevant passages, and generating responses through an externally hosted LLM. It also includes session-management functions, text interaction, optional voice input and output, and a predefined set of questions for testing.

The domain-specific design is intended to help users access information about topics such as EV batteries, charging, driving range, maintenance, and regenerative braking. Rather than requiring users to locate and read several documents manually, the

assistant is designed to return a conversational response based on retrieved content. The actual usefulness of this capability depends on retrieval quality and the accuracy of the generated response.

The prototype also provides a starting point for future development into a more accessible application. Possible improvements include a graphical web interface, integration with the main EV Management System, persistent session storage, more robust source attribution, and expanded evaluation and monitoring features.

However, the current implementation is a command-line prototype. Full integration with the overall EV Management System has not been demonstrated by the available implementation. The product should therefore be described as a functional design and implementation prototype, rather than a fully integrated or production-ready EV management service.

7.5 Limitations and threats to validity

Several limitations affect the current implementation and the conclusions that can be drawn from the research.

Knowledge-base limitations. The assistant depends on the quality, completeness, and currency of the PDF documents used as its knowledge source. Missing, outdated, or contradictory information may result in incomplete or misleading responses. The current implementation does not establish that the knowledge base covers every EV topic users may ask about.

Retrieval limitations. The system retrieves the five highest-ranked vector matches for each query. Although this provides a consistent retrieval strategy, the most relevant information may not always appear in the retrieved passages. The implementation does not demonstrate a separate reranking stage or a measured retrieval-quality threshold.

Response-generation limitations. The LLM may generate statements that are not fully supported by the retrieved context. Prompt instructions to decline unsupported or off-topic questions do not guarantee that every answer will be factually correct. The reliability of these instructions must be evaluated using representative test questions and independent checking of the generated answers.

Dependence on external services. Embedding generation, vector retrieval, and response generation depend on external services and valid API credentials. Network failures, service interruptions, rate limits, latency, and changes to service availability may affect the application's operation.

Voice-interaction limitations. Speech recognition and audio output depend on the available microphone, audio libraries, operating environment, network connectivity, and speech-recognition service. Recognition performance may vary with background noise, pronunciation, and speaking style. No measured voice-recognition accuracy has been provided.

Session persistence. Conversation history is maintained in application memory. This supports interaction within a running session but does not provide persistent storage across application restarts. A production implementation would require a suitable session-storage mechanism and appropriate data-retention controls.

User-interface and system-integration limitations. The current prototype uses a command-line interface rather than a completed graphical frontend. The available code also does not demonstrate end-to-end integration with the group's complete EV Management System. These limitations restrict conclusions about the usability and operational suitability of the product in a broader application.

Evaluation limitations. No completed quantitative test results, participant counts, questionnaire findings, or measured response-time results have been supplied for this discussion. As a result, the research cannot yet establish numerical accuracy, usability, performance, or user-satisfaction claims.

Threats to validity. The predefined test questions cover several EV topics and one unrelated topic, but this small set may not represent the diversity of real user queries. Evaluation results may also be influenced by the selected documents, the language model, prompt configuration, network conditions, and subjective reviewer judgments. A more reliable evaluation would use a larger and clearly documented test set, predefined scoring criteria, repeated response-time measurements, and independent review of factual correctness.

These limitations do not invalidate the technical implementation. Instead, they define the boundaries of the current findings and identify the additional work required before making stronger claims about the assistant's reliability, usability, and practical effectiveness.

7.6 Implications

The research has implications for EV information access, conversational-AI system development, and future EV-management applications.

Implications for EV users. A domain-focused conversational assistant may provide a more direct way to access information about charging, battery care, vehicle range, maintenance, and driving efficiency. Voice interaction may provide an alternative to typing for some users. However, the assistant should not be treated as an authoritative source for safety-critical vehicle decisions unless its information and behaviour have been appropriately validated.

Implications for system developers. The implementation illustrates the importance of treating document preparation, retrieval, response generation, and session management as interconnected components. A language model alone does not guarantee accurate domain-specific responses. Developers should evaluate retrieval quality, verify generated answers, manage external-service failures, and clearly communicate the system's limitations.

Implications for EV-management platforms. The prototype provides a foundation that could be incorporated into a broader EV Management System. Further development could connect the assistant to an accessible frontend and, where appropriate, verified system data or other project components. Such integration would require defined interfaces, access controls, data validation, and end-to-end testing.

Implications for future research. Further investigation should examine the accuracy and relevance of generated responses, the effect of retrieval settings, the reliability of off-topic handling, response time under different conditions, and user experience across text and voice interaction. Comparative testing against a

conventional FAQ system or a language-model-only baseline could help determine whether retrieval provides measurable benefits for the selected EV questions.

In summary, the research establishes a technical foundation for an EV-focused RAG virtual assistant and identifies the conditions required for further evaluation and integration. Its main implication is that combining retrieval with language generation offers a promising approach to domain-specific conversational information access, but the practical value of the approach must be demonstrated through systematic testing and integration with the intended application.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research focused on the design and development of an AI-Powered Electric Vehicle (EV) Virtual Assistant using Retrieval-Augmented Generation (RAG) with voice support. The component was developed as part of the broader research project, *Electric Vehicle Management System Using Artificial Intelligence and Machine Learning*. The main objective was to provide a conversational approach to accessing EV-related information by combining a curated knowledge base, semantic information retrieval, Large Language Model (LLM)-based response generation, conversational context, and voice interaction.

The developed prototype uses EV-related PDF documents as its knowledge source. The document content is divided into smaller overlapping text chunks, converted into vector embeddings, and stored in Pinecone. When a user submits a question, the system retrieves relevant document passages and incorporates them into a prompt for the LLM. The generated response is then returned through the command-line interface. The prototype also includes session management, predefined test questions, speech recognition, and text-to-speech functionality.

The implementation demonstrates a practical application of RAG for domain-specific conversational information access. By incorporating retrieved document content into the response-generation process, the system is designed to provide answers grounded in the available EV knowledge base. However, the reliability of these answers depends on the quality of the source documents, the relevance of the retrieved passages, and the language model's ability to generate factually consistent responses.

The prototype also demonstrates how text-based interaction can be extended through voice input and spoken responses. Nevertheless, the usability and reliability of voice interaction require further testing under realistic conditions. Similarly, although the main technical components have been implemented in code, full integration with the overall EV Management System has not been demonstrated by the available implementation.

In conclusion, the research produced a technical prototype that brings together RAG, conversational AI, session management, and voice capabilities for EV information access. The work provides a foundation for further development and evaluation. However, comprehensive testing, documented performance measurements, and integration with the main application are necessary before drawing conclusions about the system's overall effectiveness and readiness for practical deployment.

8.2 Achievement of objectives

The research objectives were established to guide the development of the EV virtual assistant. The extent to which these objectives have been addressed is summarized in Table 8.1.

Table 8.1: Achievement of Research Objectives

Research objective	**Outcome and achievement**
Develop a curated knowledge base for EV-related information.	PDF-based document loading and processing were implemented to prepare EV information for retrieval. The completeness and quality of the knowledge base require further evaluation.
Implement a RAG pipeline for retrieving relevant EV information.	Implemented using document chunking, embedding generation, Pinecone vector storage, and retrieval of the five highest-ranked matches for a query. Retrieval quality requires systematic testing.
Integrate an LLM to generate context-aware responses.	Integrated the Groq-hosted llama-3.3-70b-versatile model with retrieved context and conversational history. Response accuracy and factual consistency require empirical validation.
Develop prompt instructions and session management.	Implemented prompt instructions for EV-related questions, insufficient context, and off-topic queries, together with in-memory conversational session management. The effectiveness of these mechanisms requires testing.

Incorporate voice interaction.	Speech recognition and text-to-speech functionality were implemented in the prototype. Operational reliability and usability depend on the execution environment and require testing.
Evaluate the virtual assistant's performance and usability.	A set of eight predefined test questions and evaluation criteria were prepared. The extent of objective achievement remains incomplete until execution results and any user-validation evidence are documented.

The implementation addressed the main development objectives by establishing the knowledge-processing, retrieval, response-generation, session-management, and voice-interaction components. However, the final evaluation objective requires evidence from executed tests, measured response times, factual-consistency checks, and user feedback where applicable.

Therefore, the objectives relating to system design and implementation have been addressed at the prototype level, while comprehensive empirical validation and full integration with the overall EV Management System remain areas for further work. The final objective-achievement statements should be updated if additional testing or integration has been completed.

8.3 Individual research and product contributions

The individual research contribution is the application of established RAG and conversational-AI techniques to an EV-focused information assistant. The work combines a document-based knowledge source with semantic retrieval and LLM-based response generation, providing a structured approach to answering EV-related questions.

The main individual research contributions are as follows:

- **Domain-specific RAG implementation:** Developed a pipeline that processes EV-related PDF documents, creates vector embeddings, retrieves relevant passages, and supplies retrieved information to an LLM.

- **Context-aware response generation:** Incorporated conversational history into the response-generation workflow to support interactions involving previous messages.
- **Prompt engineering:** Defined instructions intended to focus responses on EV-related topics and handle questions for which the available context is insufficient.
- **Voice-enabled interaction:** Incorporated speech recognition and text-to-speech capabilities to support alternative methods of interacting with the assistant.
- **Evaluation planning:** Established predefined test questions and evaluation criteria for examining response relevance, factual consistency, retrieval quality, response time, and voice functionality.

The product contribution is a command-line prototype that demonstrates the interaction between the knowledge base, retrieval service, language model, session manager, and voice engine. It provides a technical foundation for developing a more accessible EV information assistant and for future integration with the broader EV Management System.

The contribution is primarily an applied software-development outcome rather than the invention of a new AI algorithm. The work demonstrates how existing technologies can be combined for an EV-focused use case. Its practical value must be established through systematic evaluation and successful integration into the intended application.

8.4 Future directions

Several improvements can be considered to enhance the accuracy, usability, reliability, and practical applicability of the EV virtual assistant.

1. Improve the knowledge base. Future development should expand and regularly update the collection of EV documents using reliable sources. Information about charging standards, battery maintenance, vehicle specifications, and driving efficiency should be reviewed to ensure that the assistant has access to relevant and

current material. Document metadata and source references should also be maintained to support traceability.

2. Improve retrieval quality. The current implementation retrieves the five highest-ranked vector matches. Future work could compare different chunk sizes, overlap values, and retrieval counts to determine their effects on answer quality. Techniques such as metadata filtering, hybrid keyword-and-vector search, and reranking could also be investigated if justified by evaluation results.

3. Strengthen factual validation and response control. Future versions could provide citations or document references with generated responses, allowing users to verify the supporting information. Additional validation mechanisms could identify unsupported claims, improve handling of insufficient context, and reduce the risk of misleading answers. These mechanisms should be tested rather than assumed to guarantee factual correctness.

4. Complete integration with the overall EV Management System. A major priority is to connect the assistant to the main application's frontend and establish clearly defined interfaces between the individual component and the other system modules. End-to-end testing should verify that users can access the assistant through the intended application and that communication between components works correctly.

5. Develop a graphical user interface. Replacing or supplementing the command-line interface with a web-based or mobile-friendly interface could improve accessibility and usability. Potential features include a chat window, conversation-history display, voice controls, loading indicators, error messages, and source references. The interface should be evaluated with representative users before conclusions are made about usability improvements.

6. Improve voice interaction. Future development could examine speech recognition under different noise conditions, support additional languages where required, and improve error handling when speech is not recognized. Voice performance should be evaluated using appropriate measures, such as transcription accuracy, successful interaction rate, and response latency.

7. Conduct comprehensive performance and user evaluation. Future evaluation should use a larger, documented test set covering different EV topics, follow-up questions, ambiguous queries, and unrelated questions. Responses should be assessed against reliable reference information. Response time should be measured across repeated trials, and user feedback should be collected where appropriate. Comparisons with a conventional FAQ system or a language-model-only baseline could help determine whether RAG provides measurable benefits for the selected use case.

8. Introduce persistent session storage and stronger security controls. Future versions could store conversation history using an appropriate database when persistence is required. Access controls, secure API-key management, data-retention policies, and protection of user information should be incorporated into the design. API credentials should be managed through environment variables or a secure secrets-management mechanism rather than being embedded directly in source code.

9. Explore integration with other EV system components. Subject to the availability of reliable data and suitable interfaces, the assistant could be extended to retrieve verified information from other modules of the EV Management System. For example, it could provide information related to charging recommendations, battery health, or vehicle range. Any personalized or real-time recommendations would require appropriate data access, validation, and safeguards.

In summary, future development should prioritize empirical validation and integration with the overall EV Management System before expanding the feature set. Improving retrieval quality, response verification, voice reliability, usability, and security would provide a stronger foundation for transforming the current prototype into a dependable EV information assistant.

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GLOSSARY

Artificial Intelligence (AI): A field of computing concerned with developing systems that perform tasks requiring capabilities such as learning, reasoning, language understanding, and decision-making.

Conversational AI: Technology that enables computer systems to communicate with users through natural-language text or speech.

Context-aware response: A response generated by considering the current question and relevant information from the preceding conversation or retrieved documents.

Document chunking: The process of dividing lengthy documents into smaller sections so that they can be processed, searched, and retrieved more efficiently.

Embedding: A numerical representation of text that captures aspects of its meaning and allows semantically related content to be identified through similarity calculations.

Hallucination: An instance in which an AI model generates information that is false, misleading, or unsupported by the available evidence.

Large Language Model (LLM): An AI model trained on large amounts of text to understand and generate human language.

Natural Language Processing (NLP): A field of AI concerned with enabling computers to process, interpret, and generate human language.

Prompt engineering: The process of designing and refining instructions provided to an AI model to guide its responses and behaviour.

Retrieval-Augmented Generation (RAG): An approach that combines information retrieval from an external knowledge source with language-model generation to produce responses informed by retrieved information.

Semantic search: A search technique that identifies information based on meaning and conceptual similarity rather than relying only on exact keyword matches.

Session management: The process of maintaining and organizing information associated with a user's interaction with an application during a conversation.

Speech recognition: Technology that converts spoken language into text that a computer system can process.

Text-to-speech: Technology that converts written text into spoken audio.

Vector database: A database designed to store and search numerical vector representations of information, often using similarity measures to retrieve relevant content.

Vector similarity: A measure of how closely two numerical vectors are related, commonly used to identify text passages that are semantically similar to a user's question.

Knowledge base: An organized collection of information used by a system to retrieve facts and support responses to user queries.

Voice user interface: An interface that allows users to interact with a system through spoken commands or verbal responses.

Factual consistency: The extent to which a generated response agrees with reliable source information and does not introduce unsupported claims.

Response latency: The time taken by a system to produce a response after receiving a user request.

Retrieval relevance: The degree to which the information retrieved by a search or retrieval process is useful for answering the user's question.

APPENDICES

Appendix A Research instruments or extended results

A.1 Purpose

This appendix provides supplementary materials used to evaluate the AI-Powered EV Virtual Assistant. It includes the predefined test questions and a template for recording the system's observed behaviour. The test results should be completed after executing the tests.

A.2 Functional Test Cases

Test ID	Test question or action	Expected outcome	Actual result
TC-01	What is an electric vehicle?	Provides a relevant explanation of an electric vehicle based on the available knowledge.	To be completed after testing
TC-02	How does a lithium-ion battery work in an EV?	Provides a relevant explanation of lithium-ion battery operation.	To be completed after testing
TC-03	What are the different types of EV charging levels?	Explains charging levels using the available EV information.	To be completed after testing
TC-04	What is the average range of electric vehicles in 2024?	Provides a contextually relevant answer and avoids unsupported generalizations.	To be completed after testing
TC-05	How do EVs compare to gasoline cars in terms of maintenance costs?	Provides a relevant comparison supported by available information.	To be completed after testing
TC-	What is regenerative	Explains regenerative braking	To be

06	braking and how does it work?	using relevant retrieved information.	completed after testing
TC-07	What is the price of crude oil today?	Follows the intended off-topic response behaviour.	To be completed after testing
TC-08	Which EV has the longest range currently available?	Provides an appropriately qualified answer supported by current, relevant evidence, if available.	To be completed after testing

A.3 Extended Evaluation Record

For each test case, record the following information:

- Test ID and date of execution
- Query submitted to the assistant
- Relevant passages retrieved from the knowledge base
- Generated response
- Response relevance and factual consistency
- Response time in seconds
- Errors or unexpected behaviour
- Pass/fail decision and justification

The evaluation should distinguish between a response that is relevant to the question and one that is factually supported by the retrieved information. Questions involving current vehicle specifications or market information require particular attention to the date and reliability of the source material.

A.4 Additional Evidence

Include screenshots of the running prototype, representative input and output examples, voice-interaction evidence, and any additional test results that were actually collected. Remove API keys, access tokens, personal information, and other confidential data before including screenshots or code extracts.

Appendix B Academic integrity and AI use disclosure

B.1 Purpose

This appendix discloses the use of AI-assisted tools during the preparation of the thesis and development-related documentation. The disclosure should be consistent with the university's academic-integrity requirements and the instructions provided for the research module.

B.2 AI-Assisted Tools and Their Uses

Tool	Purpose of use	Verification performed
ChatGPT	Assisted with drafting and refining thesis sections, organizing technical explanations, preparing test-case documentation, and improving grammar and clarity.	Reviewed the generated text for relevance to the implemented project, consistency with available evidence, and suitability for academic submission. Technical and factual claims should be checked against the implementation and cited literature.
Other AI tools, if applicable	Add the name and describe the actual use.	Describe the verification performed.

B.3 Representative Prompts and Workflows

Representative prompts used during AI-assisted thesis preparation included:

- “Draft the results and validation chapter for an AI-powered EV virtual assistant using RAG and voice support.”
- “Explain the implementation of the RAG pipeline, including document processing, embeddings, vector retrieval, and response generation.”
- “Improve the academic wording and organization of the thesis discussion and conclusion.”
- “Prepare functional test cases based on the implemented EV virtual assistant.”

These prompts are examples of the types of assistance used and should be retained only if they accurately represent the actual workflow. Add any other AI tools, prompts, or development workflows required by the university.

B.4 Verification and Student Responsibility

AI-generated content was treated as drafting and organizational assistance rather than as independent evidence of research findings. Technical descriptions should be checked against the implemented source code, and research claims should be supported by appropriate academic references. Test outcomes, numerical measurements, user feedback, and integration claims must reflect actual observations rather than generated examples.

The student remains responsible for the accuracy, originality, referencing, confidentiality, and final content of the submitted thesis. All AI use should be disclosed in accordance with the applicable institutional policy.

Appendix C: Selected Technical Specifications

C.1 Technology Stack

The prototype uses Python and the following principal technologies:

- **Document processing:** PyPDFLoader and DirectoryLoader
- **Text splitting:** RecursiveCharacterTextSplitter
- **Embedding generation:** Pinecone Inference API using llama-text-embed-v2
- **Vector storage and retrieval:** Pinecone
- **Language-model response generation:** Groq API using llama-3.3-70b-versatile
- **Conversational workflow:** LangChain and in-memory conversation history
- **Voice interaction:** SpeechRecognition, PyAudio, gTTS, pygame, pydub, and pyttsx3

C.2 Configuration and Security

The application requires appropriate API credentials and service configuration. Credentials must be supplied securely and must not be included in the submitted thesis, screenshots, or publicly shared source code. Configuration values should be documented without exposing secrets.